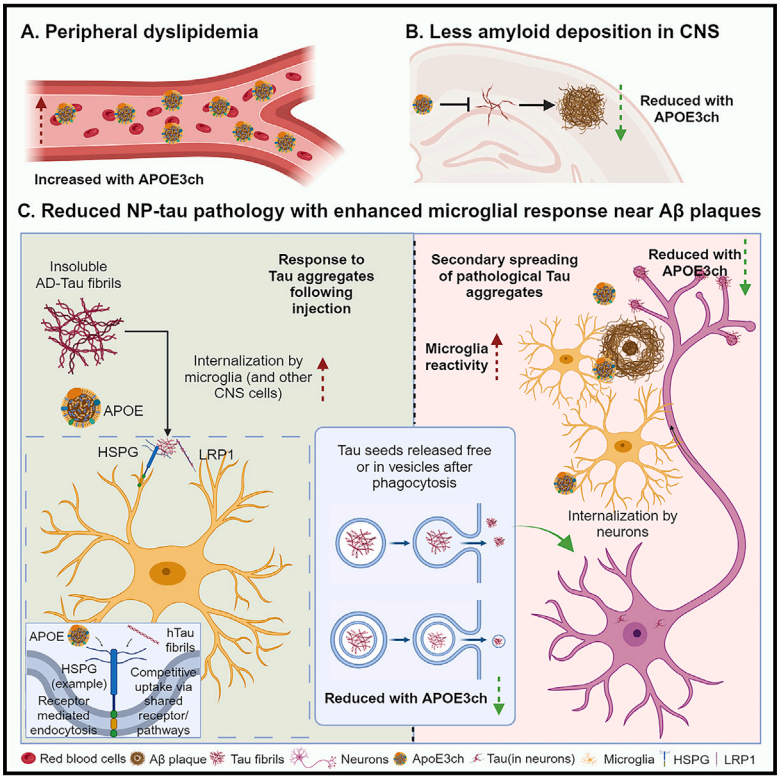
Cell

APOE3ch alters microglial response and suppresses Aβ-induced tau seeding and spread

Graphical abstract



Highlights

- APOE3ch suppresses Aβ pathology and Aβ-induced tau seeding and spreading *in vivo*
- APOE3ch enhances microglial lysosomal activity and phagocytosis of hTau fibrils
- The competitive binding of APOE over hTau to HSPG/LRP1 is weakened by APOE3ch
- Less hTau-containing material is secreted by APOE3ch than APOE3 myeloid cells

Authors

Yun Chen, Sihui Song, Samira Parhizkar, ..., Marco Colonna, Jason D. Ulrich, David M. Holtzman

Correspondence

holtzman@wustl.edu

In brief

APOE3ch reduces amyloid deposition, enhances microglial reactivity, and protects against amyloid-β-induced tau seeding and spreading *in vivo*, which is linked with increased myeloid cell tau uptake and degradation through APOE receptors.

Chen et al., 2024, Cell *187*, 428–445
January 18, 2024 © 2023 Elsevier Inc.
https://doi.org/10.1016/j.cell.2023.11.029

CellPress

CellPress

Cell

Article

APOE3ch alters microglial response and suppresses Aβ-induced tau seeding and spread

Yun Chen,[1,2] Sihui Song,[1] Samira Parhizkar,[1,3,4] Jennifer Lord,[1] Yiyang Zhu,[1] Michael R. Strickland,[1] Chanung Wang,[1] Jiyu Park,[1] G. Travis Tabor,[1] Hong Jiang,[1,3,4] Kevin Li,[1] Albert A. Davis,[1,3] Carla M. Yuede,[1,3,4,5,6] Marco Colonna,[2,3] Jason D. Ulrich,[1,3,4] and David M. Holtzman[1,3,4,7,*]

[1]Department of Neurology, Washington University School of Medicine, St. Louis, MO 63110, USA
[2]Department of Pathology and Immunology, Washington University School of Medicine, St. Louis, MO 63110, USA
[3]Hope Center for Neurological Disorders, Washington University School of Medicine, St. Louis, MO 63110, USA
[4]Knight Alzheimer's Disease Research Center, Washington University School of Medicine, St. Louis, MO 63110, USA
[5]Department of Psychiatry, Washington University School of Medicine, St. Louis, MO 63110, USA
[6]Department of Neuroscience, Washington University School of Medicine, St. Louis, MO 63110, USA
[7]Lead contact
*Correspondence: holtzman@wustl.edu
https://doi.org/10.1016/j.cell.2023.11.029

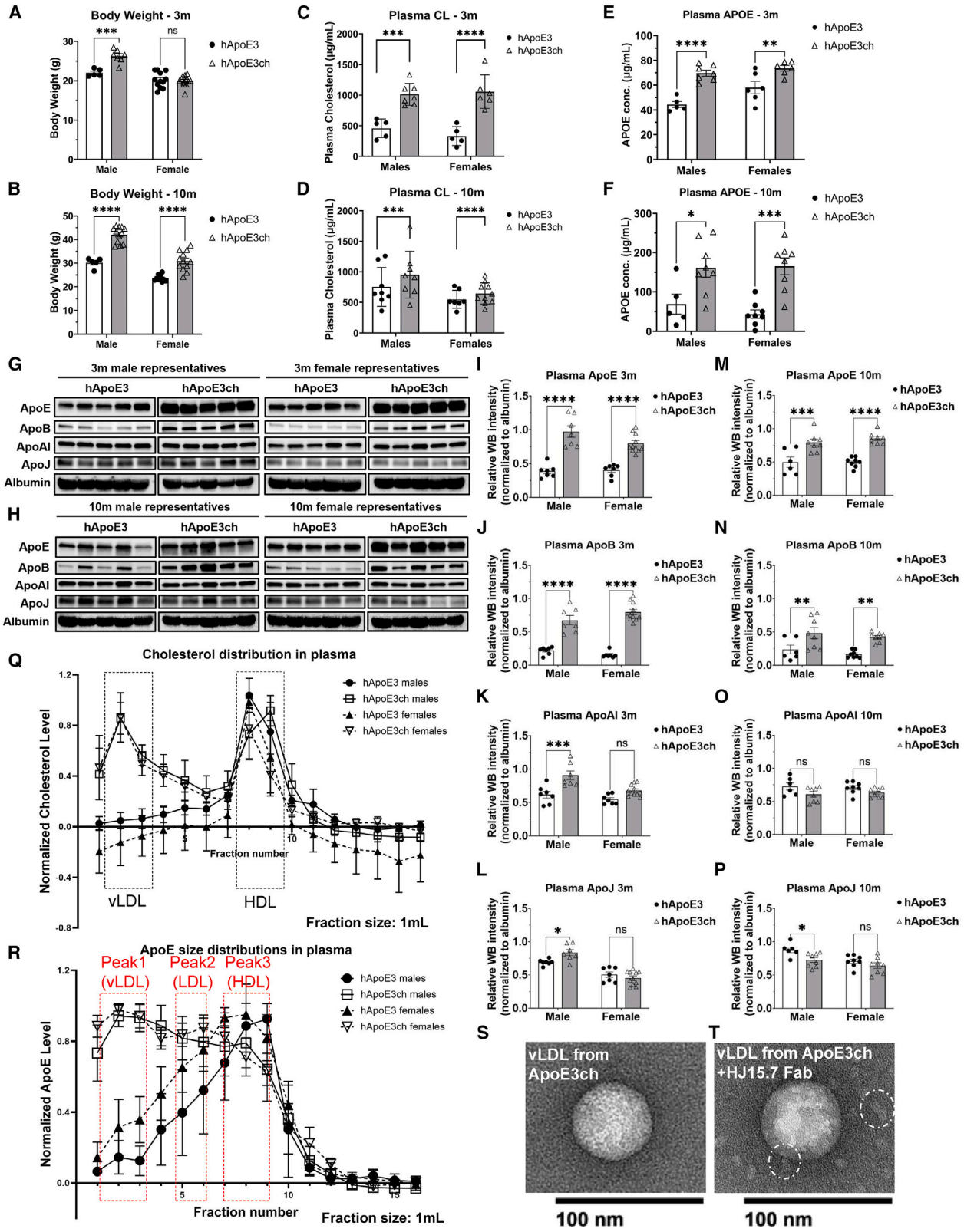
SUMMARY

A recent case report described an individual who was a homozygous carrier of the *APOE3* Christchurch (*APOE3ch*) mutation and resistant to autosomal dominant Alzheimer's Disease (AD) caused by a *PSEN1-E280A* mutation. Whether *APOE3ch* contributed to the protective effect remains unclear. We generated a humanized *APOE3ch* knock-in mouse and crossed it to an amyloid-β (Aβ) plaque-depositing model. We injected AD-tau brain extract to investigate tau seeding and spreading in the presence or absence of amyloid. Similar to the case report, *APOE3ch* expression resulted in peripheral dyslipidemia and a marked reduction in plaque-associated tau pathology. Additionally, we observed decreased amyloid response and enhanced microglial response around plaques. We also demonstrate increased myeloid cell phagocytosis and degradation of tau aggregates linked to weaker APOE3ch binding to heparin sulfate proteoglycans. *APOE3ch* influences the microglial response to Aβ plaques, which suppresses Aβ-induced tau seeding and spreading. The results reveal new possibilities to target Aβ-induced tauopathy.

INTRODUCTION

Alzheimer's disease (AD) is the most common cause of dementia and can be categorized as early-onset AD (clinical onset before age 65) and late-onset AD (LOAD) (clinical onset age 65 and over).[1] All types of AD share classic pathological hallmarks including extracellular amyloid-β (Aβ) deposition in neuritic plaques and intraneuronal aggregation of hyperphosphorylated tau into neurofibrillary tangles (NFTs).[2] For both early- and late-onset AD, Aβ plaques begin to form approximately 20 years prior to the onset of cognitive decline, which coincides with fibrillar tau pathology and neurodegeneration in the neocortex.[3] Approximately 1%–2% of AD cases are classified as autosomal dominant Alzheimer's disease (ADAD), which is caused by mutations in one of 3 genes: *APP*, *PSEN1*, or *PSEN2*.[4] Almost all of these individuals develop early-onset AD. The age of onset of clinical disease in ADAD is strongly linked to the specific gene mutation and likely other factors.[5]

The *APOE* gene strongly influences the risk of LOAD, with the *APOE4* allele increasing AD risk and the *APOE2* allele decreasing AD risk relative to the most common *APOE3* allele.[6] In 2019, a case report from the world's largest ADAD kindred revealed the first potential protective mutation against ADAD, which was a homozygous variant in two *APOE3* alleles—R136S—called APOE3-Christchurch (APOE3ch).[7] The APOE3ch homozygous carrier with the PSEN1-E280A mutation developed very mild cognitive decline in her 70's, approximately 30 years after the expected age of onset associated with the PSEN1-E280A mutation.[7] The individual in her 70's had a higher level of Aβ pathology than PSEN1-E280A carriers assessed in their mid-40s, at the typical age of onset of cognitive decline. Interestingly, the amount of tauopathy in regions typically affected in AD, such as the temporal and parietal cortex, was very limited. Subsequent postmortem studies verified a disconnection between elevated Aβ burden and low tau pathology and neurodegeneration.[8] Although the potential protective effects of APOE3ch are an exciting observation, the rarity of the APOE3ch variant and isolated case-report nature of the initial observation limit the ability to attribute the phenotype observed to the APOE3ch variant or understand potential protective mechanisms.



(legend on next page)

The Christchurch R136S variant was first discovered in APOE2 carriers with type 3 hyperlipoproteinemia (HLP-III).[9] APOE-associated dyslipidemia is linked with molecular changes in the receptor binding site of APOE2 that impair receptor-mediated lipoprotein clearance by the liver.[10] Many of the APOE2 variants that caused HLP-III impair receptor binding to LDLR, HSPG, or LRP1.[10,11] Mutagenesis of R136S on an APOE3 background reduced APOE binding to the LDLR and to heparin sulfate.[12,13] Indeed, in the case of the APOE3ch homozygous carrier with the PSEN1-E280A mutation,[7] HLP-III was diagnosed according to the patient's plasma measurements.[7] Thus, similar reductions in ligand-receptor interactions between APOE3ch and the LOAD protective variant APOE2 can manifest similar plasma lipoprotein and dyslipidemia phenotypes.

To further understand whether and how the APOE3ch variant could impact AD pathology, we generated a humanized knock-in (KI) mouse expressing human APOE3-R136S and compared this to APOE3 KI mice. To understand whether the APOE3ch variant influenced Aβ deposition and Aβ-induced tau seeding and spreading, we crossed APOE3ch KI and APOE3 KI mice with an Aβ-depositing model, APPPS1-21(referred as APP/PS1).[14] We injected insoluble tau fibrils derived from human AD brain extract (AD-tau) to assess Aβ-induced tau seeding and spreading. We aimed to test whether APOE3ch would affect Aβ pathology or early tau seeding and spreading induced by amyloid deposition. Interestingly, we found that APOE3ch reduced overall Aβ plaque deposition and markedly reduced peri-amyloid plaque tau seeding and spreading. We found an enhanced peri-plaque microglial response and increased myeloid cell phagocytosis and degradation of human tau fibrils in the presence of APOE3ch. Enhanced tau uptake by myeloid cells was linked with reduced binding of APOE3ch to HSPG and LRP1 *in vitro*. Further, we showed that the APOE3ch-expressing myeloid cells secreted less previously phagocytosed tau compared to APOE3-expressing cells, which potentially contributes to the reduced tau spreading in the injection model. The data suggest that APOE3ch impedes amyloid-induced tau seeding and spreading via an effect on microglia.

RESULTS

Peripheral dyslipidemia and related APOE-associated phenotype changes by APOE3ch

The previous case report of an APOE3ch homozygous individual with the PSEN1-E280A mutation reported a peripheral dyslipidemia phenotype, consistent with previous findings of the effect of R136S on receptor binding.[7,10,12] To investigate whether homozygous R136S on an APOE3 background is sufficient to induce dyslipidemia, we assessed plasma and CNS lipoproteins and associated pathways in 3-month-old and 10-month-old APOE3 and APOE3ch homozygous KI mice (Figures 1 and 2). At 3 months of age, male, but not female, APOE3ch mice exhibited a significant increase of body weight (Figure 1A), but by 10 months of age, both sexes had increased body weight (Figure 1B). Both male and female APOE3ch mice had elevated total plasma cholesterol compared to APOE3 at 3 and 10 months (Figures 1C and 1D). Utilizing an anti-human APOE antibody that recognizes the APOE C-terminal region, we found significantly elevated plasma APOE protein levels at both ages (Figures 1E and 1F). Western blot analysis confirmed increased plasma APOE and APOB protein levels in male and female APOE3ch mice, while two other lipoproteins (APOA-I and APOJ) were increased only in male APOE3ch mice at 3 months (Figures 1G, S1A, and 1I–1L). Interestingly, the increase of APOE and APOB persisted to the age of 10 months, while APOA-I and APOJ showed a trend toward reduction in both sexes (Figures 1H, S1B, and 1M–1P). To further investigate the consequence of plasma APOE changes, we utilized size-exclusion chromatography (SEC) to separate plasma lipoprotein components from 3-month-old mice based on size. Both male and female APOE3ch mice had very low-density lipoprotein particles (vLDL) that were not evident in APOE3 mice (Figure 1Q, individual

Figure 1. Type 3 hyperlipoproteinemia (HLP-III) like dyslipidemia phenotype caused by APOE3ch

(A and B) Body weights of APOE3 vs. APOE3ch mice at 3 months (m) of age (A) and 10m (B). N = 5/7, 3m APOE3 males/females; N = 12/12, 3m APOE3ch males/females; N = 5/10, 10m APOE3 males/females; N = 12/12, 10m APOE3ch males/females.

(C and D) Total cholesterol (μg/mL) in the plasma of APOE3 vs. APOE3ch mice at 3m (C) and 10m (D). N = 5/5, 3m APOE3 males/females; N = 7/6, 3m APOE3ch males/females; N = 8/7, 10m APOE3 males/females; N = 8/9, 10m APOE3ch males/females.

(E and F) Total APOE concentrations (μg/mL) in the plasma of APOE3 vs. APOE3ch mice at 3m (E) and 10m (F). N = 5/6, 3m APOE3 males/females; N = 7/6, 3m APOE3ch males/females; N = 5/8, 10m APOE3 males/females; N = 8/8, 10m APOE3ch males/females.

(G and H) Representative Western blot images of major lipoproteins in the plasma samples from 3m (G) and 10m (H). All individual bands are shown in Figures S1A and S1B.

(I–P) Quantification of the Western blot intensity of the plasma lipoproteins normalized to albumin (Figures S1A and S1B) in the 3m and 10m plasma samples. N = 7/7, 3m APOE3 males/females; N = 7/11, 3m APOE3ch males/females; N = 6/8, 10m APOE3 males/females; N = 8/8, 10m APOE3ch males/females.

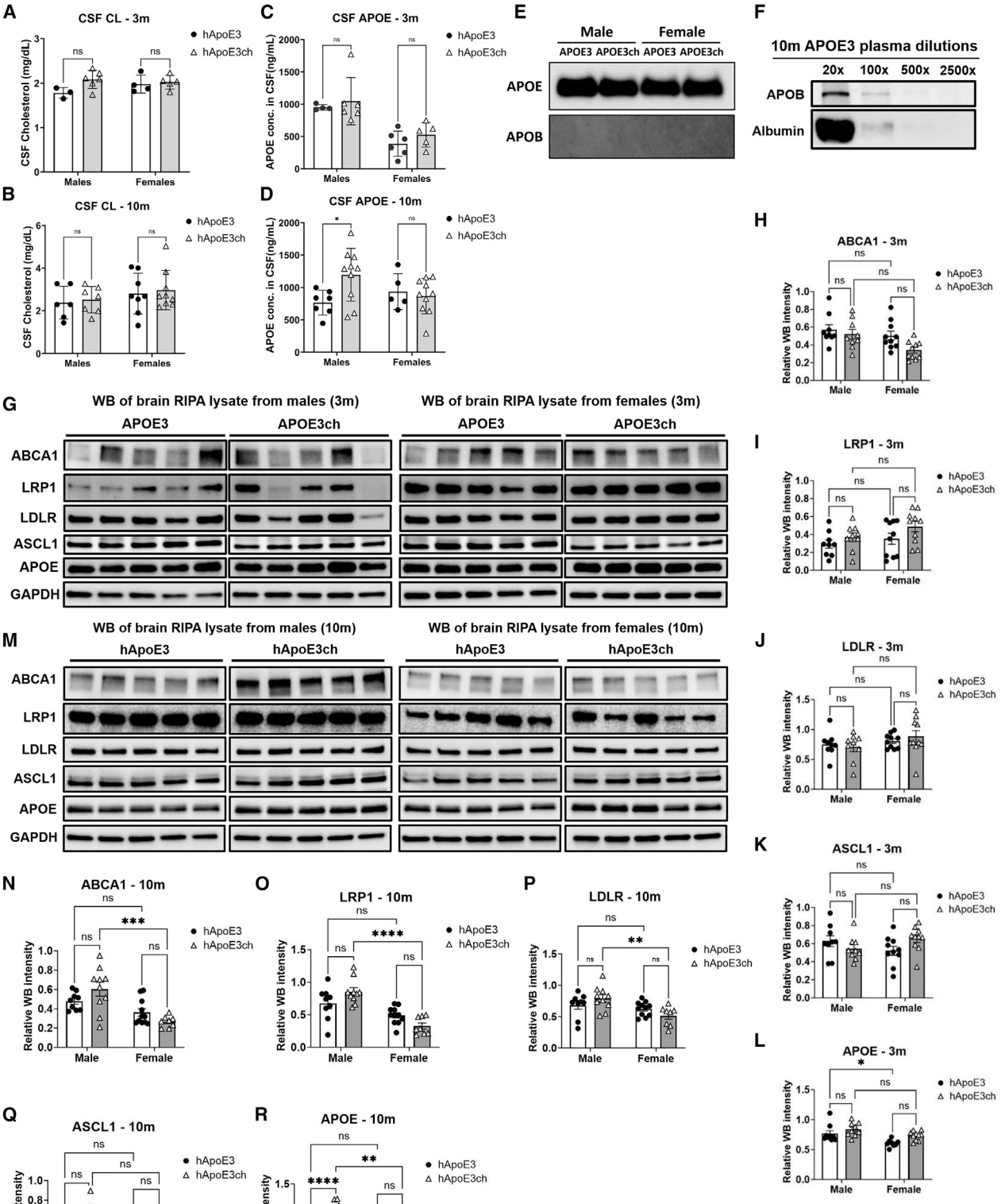
(Q) Normalized cholesterol levels in each SEC-separated plasma fractions from APOE3 vs. APOE3ch mice (N = 5/6, 3m APOE3 males/females; N = 7/5, 3m APOE3ch males/females). The vLDL and HDL lipoparticles were highlighted in the dash box according to the standard curve (methods). The cholesterol level in each fraction has been normalized to the average cholesterol level detected in that individual (Figures S1C and S1D).

(R) Normalized APOE level by ELISA measurements in each SEC-separated fraction from APOE3 vs. APOE3ch mice (N = 5/6, 3m APOE3 males/females; N = 7/5, 3m APOE3ch males/females). The vLDL, LDL and HDL lipoparticles highlighted in the dash box according to the standard curve (methods). The APOE level in each fraction is normalized to the average APOE level detected in that individual (Figures S1E and S1F).

(S and T) Negative stain transmission electron microscopy images of vLDL particles isolated from APOE3ch plasma with or without anti-hAPOE HJ15.7 antibody Fab. A free HJ15.7 Fab and one bound to vLDL are highlighted in the white dash circles.

In all graphs, the statistical analyses were performed with two-way ANOVA and Šídák's multiple comparisons test for the adjusted p values. The error bars represent SEM values. The results designated with ns were not significant, whereas results designated with * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$) were significant according to adjusted p values in each indicated comparison.

See also Figure S1.



(legend on next page)

plots in Figures S1C and S1D). Paired with APOE ELISA, we identified high APOE levels associated with the vLDL and LDL lipoproteins specifically in the APOE3ch mice, whereas APOE3 mice only contained APOE-associated HDL particles (Figure 1R, individual plots in Figures S1E and S1F). The integrity of the vLDL lipoparticles and recognition of APOE by an anti-APOE antibody were further verified by negative stain transmission electron microscopy (NS-TEM) (Figures 1S and 1T). Thus, APOE3ch drives a dyslipidemia phenotype in the periphery.

To determine whether APOE3ch would influence CNS lipoprotein metabolism, we collected both CSF and brain samples from 3-month-old and 10-month-old APOE3 and APOE3ch mice. In contrast to the periphery, there was no difference in total CSF cholesterol levels between APOE3 and APOE3ch mice at either age (Figures 2A and 2B). CSF APOE protein levels between APOE3 and APOEch mice were generally unchanged (Figures 2C–2E), except that male APOE3ch mice at 10 months showed a significant increase of APOE. The purity of our CSF samples from blood contamination could be validated by undetectable levels of APOB (Figures 2E and 2F). We also observed a modest but statistically significant increase of APOE protein in brain lysates of male 10-month-old, but not 3-month-old, APOE3ch mice (Figures 2G–2Q). Additionally, we did not observe changes in the levels of APOE-related proteins, including the lipid transporter ABCA1, APOE receptors LDLR and LRP1, or the lipid pathway regulator ASCL1 between APOE3 and APOE3ch (Figures 2G–2Q). Therefore, we conclude that APOE3ch does not appear to cause dramatic changes in the level of lipoproteins or lipoprotein-related proteins in the CNS. We consistently observed sex differences between 10-month-old APOE3ch males and females in terms of lipid-associated proteins, consistent with previous literature of sex differences in lipid metabolism.[15] Finally, there were no significant differences in memory-related behavioral tests between APOE3 and APOE3ch mice at 8–10 months of age (Figure S2).

Total amyloid burden and NP-tau pathology are reduced in APP/PS1:APOE3ch

We next addressed two major questions regarding the potential protective effect of APOE3ch: whether the APOE3ch mutation could affect Aβ deposition and whether APOE3ch affected amyloid-dependent tau seeding and spreading. We first crossed APOE3 and APOE3ch mice to APP/PS1 mice that develop Aβ deposition and assessed amyloid pathology at 6 months of age. As the APOE3ch homozygous carrier with the PSEN1-E280A mutation was female, and there might be a sex-dependent lipoprotein pathway adaptation between males and females (Figure 2), we chose to study female mice first. We observed a modest but statistically significant reduction of $X34^+$ fibrillar Aβ plaques and reduced average fibrillar plaque size in APP/PS1:APOE3ch mice compared to APP/PS1:APOE3 at 6 months (Figures 3A–3C). However, there was no significant difference in the number of $X34^+$ or Aβ-immunoreactive plaques nor total Aβ plaque burden immunostained with an N-terminal anti-Aβ antibody (Figures S3A–S3E). Consistent with the modest decrease in fibrillar plaque burden, but no decrease in Aβ staining, we observed a trend, but not a statistically significant difference, in insoluble Aβ42 and APOE in APP/PS1:APOE3ch mice (Figures S3F–S3I). We also observed a significant increase of soluble Aβ40 in APP/PS1:APOE3ch mice (Figure S3F), but notably, there was no significant difference in the level of cortical PBS-soluble APOE between APP/PS1:APOE3ch and APP/PS1:APOE3 mice (Figure S3H). We also did not observe any statistical differences in memory-associated behavioral tests between APP/PS1:APOE3 and APP/PS1:APOE3ch mice at 4–6 months of age (Figures S2A–S2H). Of note, the APP/PS1 mice at this age did not differ from the non-APP/PS1 mice in the behavioral tests assessed.

We next used human AD-tau injection to assess both cerebral amyloidosis and neuritic plaque-associated tau pathology (NP-tau pathology), as described previously.[16–18] Female APP/PS1:APOE3 and APP/PS1:APOE3ch KI mice received unilateral intra-parenchymal injections of human AD-tau at 6 months of age into one cortical and one hippocampal site and were sacrificed 3.5 months post injection (Figure 3D).[17] In this model, endogenous mouse tau is induced to become hyperphosphorylated and aggregated following injection of human AD-tau, in which the AD tauopathy-associated epitope MTBR-tau243 could be detected (Figure S3J).[19] The seeded tau pathology occurs around the amyloid plaques in dystrophic neurites and additionally spreads to synaptically-connected brain regions, both ipsilaterally and contralaterally to the side of injection.[17] At 9.5 months of age, AD-tau-injected APP/PS1:APOE3ch mice exhibited an ∼40% reduction in $X34^+$ fibrillar Aβ plaques, decreased average plaque size and a decrease in total Aβ plaque burden and number of plaques (Figures 3E–3L). As described previously, we also found that recombinant human APOE3ch leads to reduced maximal Aβ42 fibril formation compared to

Figure 2. Assessment of APOE and lipid-associated proteins in the CNS of APOE3 and APOE3ch mice

(A and B) Total cholesterol (mg/dL) in the CSF of APOE3 vs. APOE3ch mice at 3m (A) and 10m (B). N = 3/4, 3m APOE3 males/females; N = 6/5, 3m APOE3ch males/females; N = 6/8, 10m APOE3 males/females; N = 7/9, 10m APOE3ch males/females.

(C and D) Total APOE concentrations (ng/mL) in the CSF of APOE3 vs. APOE3ch mice at 3m (C) and 10m (D). N = 4/6, 3m APOE3 males/females; N = 6/5, 3m APOE3ch males/females; N = 7/5, 10m APOE3 males/females; N = 10/10, 10m APOE3ch males/females.

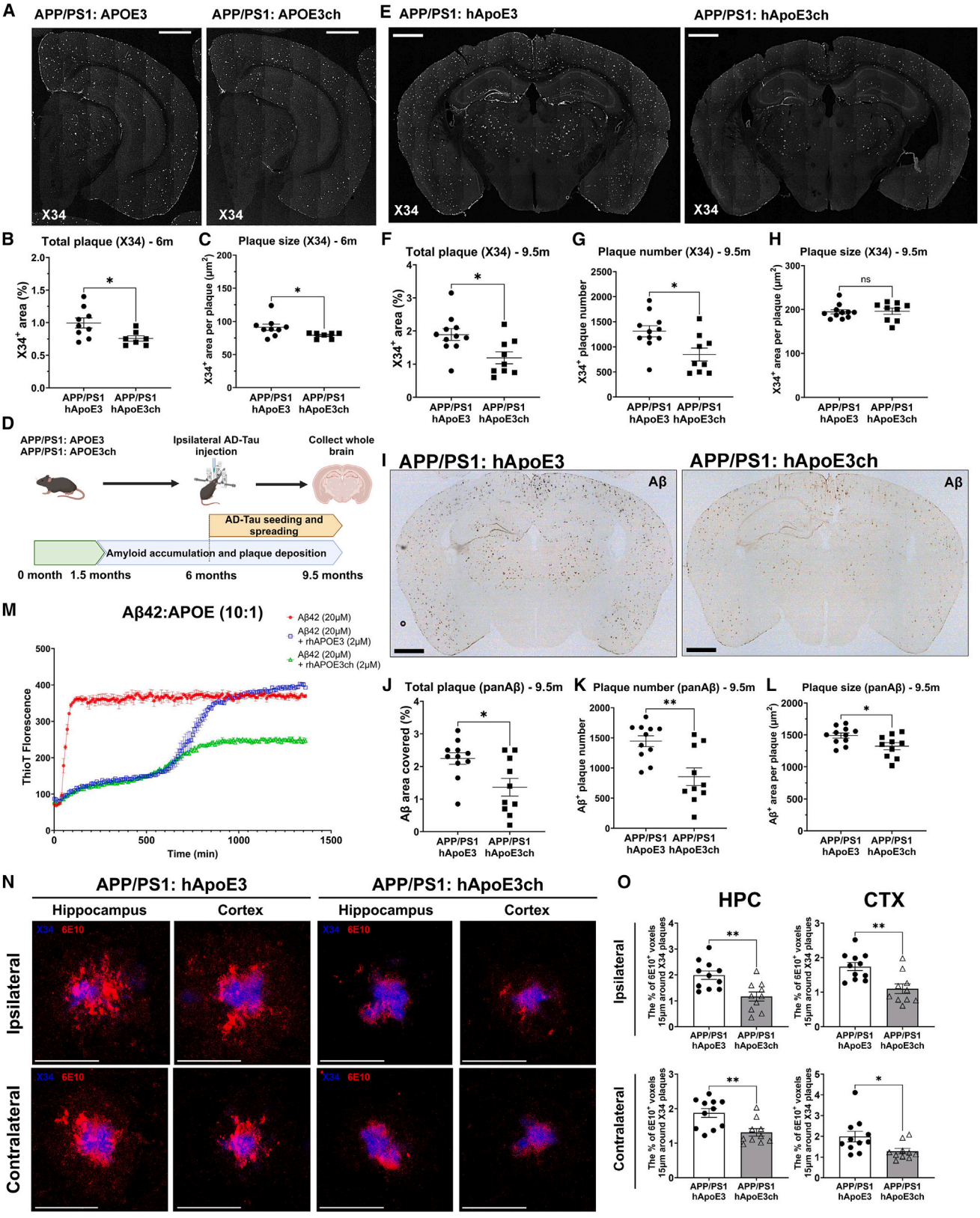
(E and F) Western blot image of APOE, and APOB lipoproteins in the pooled CSF samples from fig.2A-D (E). APOB control WB of plasma serial dilutions (F).

(G–L) Western blot images of APOE and APOE-related proteins (labeled) in brain extractions from 3m cohort along with quantification of results. All individual bands were shown in Figure S1G. N = 9/10, 3m APOE3 males/females; N = 9/10, 3m APOE3ch males/females.

(M–R) Western blot images of APOE and APOE-related proteins (labeled) in brain extractions from 10m cohort along with quantification of results. All individual bands were shown in Figure S1H. N = 9/11, 10m APOE3 males/females; N = 10/8, 10m APOE3ch males/females.

In all graphs, the statistical analyses were performed with two-way ANOVA and Šídák's multiple comparisons test for the adjusted p values. The error bars represent SEM values. The results designated with ns were not significant ($p > 0.05$), whereas results designated with * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$) were significant according to adjusted p values in each indicated comparison.

See also Figures S1 and S2.



(legend on next page)

APOE3 using an *in vitro* ThioT assay (Figure 3M).[7,20] This effect of APOE3ch vs. APOE3 on reducing maximal Aβ fibril formation *in vitro* under the conditions we utilized suggests one possible reason for why there were decreased Aβ fibrils found *in vivo* in the APP/PS1:APOE3ch vs. APOE3 mice. Indeed, APP/PS1:APOE3ch mice formed more small amyloid plaques (area <750 μm^2) detected by Aβ staining and to a lesser extent in $X34^+$ fibrillar plaque-stained cores (core area <120 μm^2) (Figures S3K–S3P). We further performed high power confocal imaging and analyzed the total Aβ detected by an anti-Aβ antibody (6E10) together with $X34^+$ staining (Figures 3N, 3O and S3Q–S3S). No difference of 6E10% around $X34^+$ plaques was detected between APP/PS1:APOE3 and APP/PS1:APOE3ch mice at 6 months (Figures S3Q–S3S). However, the 6E10-detectable Aβ around individual $X34^+$ plaques in APP/PS1:APOE3ch mice was significantly reduced compared to APP/PS1:APOE3 mice in AD-tau-injected 9.5-month-old mice (Figures 3N and 3O).

To investigate the effects of APOE3ch on NP-tau pathology, we assessed NP-tau throughout the hippocampus, somatosensory cortex, and entorhinal cortex with the anti-phospho-tau antibody AT8 that recognizes tau phosphorylated at positions Ser202/Ser205 (Figure 4A). We observed an overall reduction of total NP-tau in the ipsilateral and contralateral hippocampus and cortex (Figures 4A, 4B, S4A and S4B) in APP/PS1:APOE3ch vs. APP/PS1:APOE3 mice. Total NP-tau pathology was significantly higher in the ipsilateral dentate gyrus of APP/PS1:APOE3 than APP/PS1:APOE3ch, indicating attenuated AD-tau seeding in the presence of APOE3ch. We also observed a reduction in AD-tau spreading to secondary pathological sites including the contralateral hippocampus, CA1, somatosensory cortex and entorhinal cortex in the APP/PS1:APOE3ch vs. the APP/PS1:APOE3 mice (Figures 4A, 4B, S4A and S4B). Given that NP-tau pathology is driven by Aβ plaques,[16,17] we conclude that APOE3ch expression resulted in less AD-tau seeding and spreading to different brain regions, possibly in part due to reduced Aβ burden compared to mice expressing APOE3.

Differences in amyloid-dependent tau seeding and spreading can arise from differences in the microenvironment around individual amyloid plaques.[17,18,21] Therefore, we performed confocal analysis to assess NP-tau pathology on a per-plaque level (Figures 4C and 4D), a pathology that is not evident in non-inoculated APP/PS1 mice (Figures S4C–S4E). We observed a marked decrease of $AT8^+$ puncta around individual $X34^+$ amyloid plaques in the APP/PS1:APOE3ch mice, both in the ipsilateral and contralateral hippocampal and cortical regions (Figures 4C and 4D), indicating reduced NP-tau aggregation per plaque. This reduction of $AT8^+$ volume around $X34^+$ plaques was not correlated with fibrillar plaque size (Figures S3K–S3P).[20,21] Since NP-tau aggregation occurs within plaque-associated dystrophic neurites, we assessed the peri-plaque staining of BACE1, a protein that accumulates in dystrophic neurites. We observed significantly reduced $BACE1^+$ dystrophic neurites around individual $X34^+$ plaques both in the ipsilateral and contralateral cortex and hippocampus of APP/PS1:APOE3ch mice (Figures 4E and 4F). Another pathological alteration around amyloid plaques is a loss of synapses.[22,23] We assessed synapsin and PSD95 puncta around amyloid plaques in APP/PS1:APOE3 and APP/PS1:APOE3ch mice. We observed increased synapsin staining around $X34^+$ plaques in brain regions where tau-seeding and spreading was evident in APP/PS1:APOE3ch compared to APP/PS1:APOE3 mice (Figures 4G and 4H). PSD95 staining was also increased around plaques in APP/PS1:APOE3ch vs. APP/PS1:APOE3 mice at the ipsilateral sites, although there was no statistical difference in contralateral regions (Figures S4F and S4G). The protective effect of ApoE3ch on dystrophic neurites and synaptic markers around Aβ plaques were seen in the tau seeding model but not in the non-inoculated, 6-month-old APP/PS1 mice (Figures S4H–S4N). Together, these data support the conclusion that APOE3ch reduces overall NP-tau pathology as well as NP-tau seeding and spreading on a per plaque basis. In addition, APOE3ch results in less overall damage to neurites and synapses around amyloid plaques when p-tau pathology is present.

Microglial reactivity is enhanced specifically around plaques in APP/PS1:APOE3ch mice

Because we previously reported that microglia influence Aβ plaque-associated tauopathy,[17,18] we hypothesized that the strongly reduced NP-tau seeding and spreading may be due to differences in microglial response around plaques because of APOE3ch. We performed immunostaining with well-characterized markers for assessing microglial reactivity, including APOE, CD68, TMEM119, and CLEC7A in the APP/PS1 model

Figure 3. Reduced amyloid pathology in the APP/PS1:APOE3ch mice

(A–C) Representative confocal images of X34 staining (white) of fibrillar Aβ plaques in APP/PS1:APOE3 and APP/PS1:APOE3ch mice (6m) and quantifications as labeled. Scale bar in A, 1000μm. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch.

(D) Scheme of experimental timeline and pathological progress of the human AD-tau injection into APP/PS1:APOE mouse models.

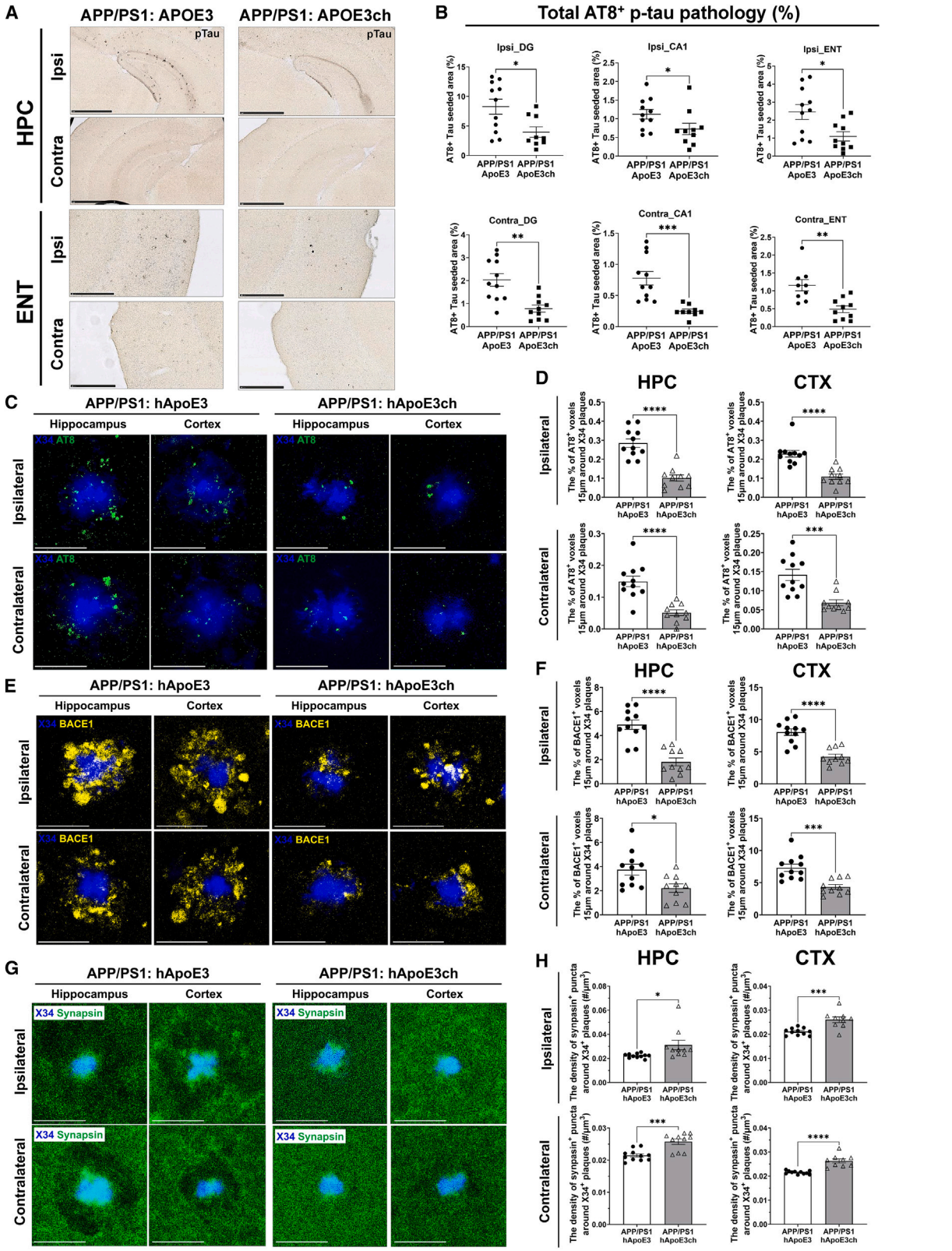
(E–H) Representative confocal images of X34 staining (white) of fibrillar Aβ plaques in the whole brain of AD-tau-seeded APP/PS1:APOE3 and APOE3ch mice (9.5m) and quantifications as labeled. Scale bars in E, 1000μm. N = 11, APP/PS1:APOE3; N = 9, APP/PS1:APOE3ch.

(I–L) Representative images of anti-Aβ (HJ3.4) DAB staining of total Aβ in AD-tau-seeded APP/PS1:APOE3 and APP/PS1:APOE3ch and quantification. Scale bars in I, 1000μm. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(M) The Aβ42 fibrillization ThioT curve with recombinant human APOE3 and APOE3ch at concentrations as labeled. Each dot represents an average signal from independent technical triplicates at one time point as indicated.

(N and O) Representative confocal images of 6E10-647 (anti-Aβ) staining (red) with $X34^+$ plaques (blue) in the ipsilateral and contralateral non-Tau-seeded cortex from AD-tau-seeded APP/PS1:APOE3 and APP/PS1:APOE3ch mice and quantification of the percent of $6E10^+$ voxels within 15μm around $X34^+$ plaques. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch. Scale bars in N are 30 μm.

In all column graphs, the statistical analyses were performed with student t-tests for mean values from individual animals. The error bars represent SEM. The results designated with ns were not significant, whereas results designated with * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$) were significant. See also Figures S2 and S3 and Table S1.



(legend on next page)

with and without AD-tau inoculation (Figures 5 and S5). Consistent with reduced Aβ plaque burden, we observed a reduction in global microgliosis in the ipsilateral and contralateral hippocampus and cortex of AD-tau-inoculated APP/PS1:APOE3ch vs. APP/PS1:APOE3 mice (Figures S5A–S5E). Overall, APOE staining was also reduced in the APP/PS1:APOE3ch cortex and hippocampus (Figures S5F–S5I). The percentage of APOE$^+$ microglia was unchanged between APP/PS1:APOE3 and APP/PS1:APOE3ch except in the contralateral cortex (Figures S5J–S5M).

Interestingly, we observed that microglial clustering around each X34$^+$ plaque was significantly increased in the APP/PS1:APOE3ch mice across all ipsilateral and contralateral regions (Figures 5A and 5B). Moreover, CD68$^+$ microglial phago-lysosome reactivity around Aβ plaques was strongly enhanced (Figures 5A and 5C). Enhanced microglial clustering around amyloid plaques and CD68 and APOE staining were also evident at 6 months of age (Figures 5D–5H). Additionally, in AT8$^-$ regions rostral to the injection site in the AD-tau-injected cohort, we found a similar increase in CD68 staining in plaque-associated microglia in APP/PS1:APOE3ch mice (Figures S5N–S5P), although there was no significant difference in overall microglial clustering or microglial APOE staining (Figures S5Q–S5U). However, we did not detect a difference in another disease-associated microglia (DAM)-associated marker, CLEC7A, around Aβ$^+$ plaques in APP/PS1:APOE3ch vs. APP/PS1:APOE3 mice at 6 months or in AD-tau injected 9.5-month-old mice (Figures S5V–S5AC). Together, these differences indicate that APOE3ch altered microglial phagolysosomal reactivity toward AD pathology. In conjunction with this change, staining for the homeostatic-associated microglial marker, TMEM119, was significantly reduced around each plaque in AD-tau-injected 9.5-month-old APP/PS1:APOE3ch mice compared with the APP/PS1:APOE3 (Figures 5I and 5J). We did not observe statistically significant changes in TMEM119 in microglia between APP/PS1:APOE3 and APP/PS1:APOE3ch around plaques at 6 months of age (Figures S5AD–S5AF). We conclude that in the APP/PS1:APOE3ch mice microglial reactivity is altered around Aβ plaques, particularly in the presence of NP-tau pathology.

Of note, GFAP$^+$ astrocytes were reduced in different brain regions in the AD-tau-seeded APP/PS1:APOE3ch mice (Figures S6A and S6B). We also observed a trend but no significant reduction of total GFAP$^+$ astrocytes in the APP/PS1:APOE3ch mice (Figures S6C and S6D), but no changes in the GFAP$^+$ astrocytes around each plaque (Figures S6E and S6F). Given that no difference in GFAP$^+$ astrocytes was observed in non-APP/PS1 mice that are APOE3 vs. APOE3ch mice at 10 months of age (Figures S6G and S6H), the difference in GFAP$^+$ astrocytes likely stems from the overall reduced amyloid burden in the APP/PS1:APOE3ch mice (Figures S6A–S6D).

The protection against AD-tau seeding and propagation is not evident in APOE3ch mice without amyloidosis

To further investigate whether APOE3ch affected tau seeding and propagation in the absence of amyloid pathology, we performed AD-tau injection into 9.5-month-old APOE3 and APOE3ch mice and assessed the propagation of AT8-tau pathology across brain regions 6 months later (Figure 6A). In previous studies, AD tau injection into the hippocampus of mice without amyloid[16] resulted in mouse tau pathology that preferentially formed neurofibrillary tangles (NFTs) and p-tau neuropil staining and propagated through connected neuronal circuits to certain regions including ventral retrosplenial cortex (RSPv), entorhinal cortex (ENT) and hypothalamus (HY) (Figure 6B). Similarly,[16,24] we observed AT8$^+$ tau pathology surrounding the injection site as well as distant from the original injection sites in the APOE KI mice (Figures 6B and 6C). Interestingly, we did not observe any difference in total AT8 pathology across 8 brain regions including the primary injection site (ipsilateral dentate gyrus, DG) and 7 propagation sites (ipsilateral RSPv, ENT, and HY and contralateral DG, RSPv, ENT, and HY) between the APOE3 and APOE3ch mice (Figures 6D–6G). However, despite the limited differences in tau propagation in this model, we observed an increase of total CD68 in microglia of APOE3ch in regions with high AT8$^+$ staining where intraneuronal pTau aggregates were evident in the DG and RSPv, but not in the ENT (Figures 6H and 6I). Overall, these results suggest that APOE3ch enhanced microglial reactivity but did not prevent tau propagation in the absence of amyloid pathology.

APOE3ch myeloid cells specifically phagocytose more hTau fibrils and release less tau

To investigate the altered APOE3ch microglial response and its impact on amyloid and NP-tau pathology, we conjugated a

Figure 4. Reduced Tau seeding and spreading and neuronal dystrophy in the APP/PS1: APOE3ch mice

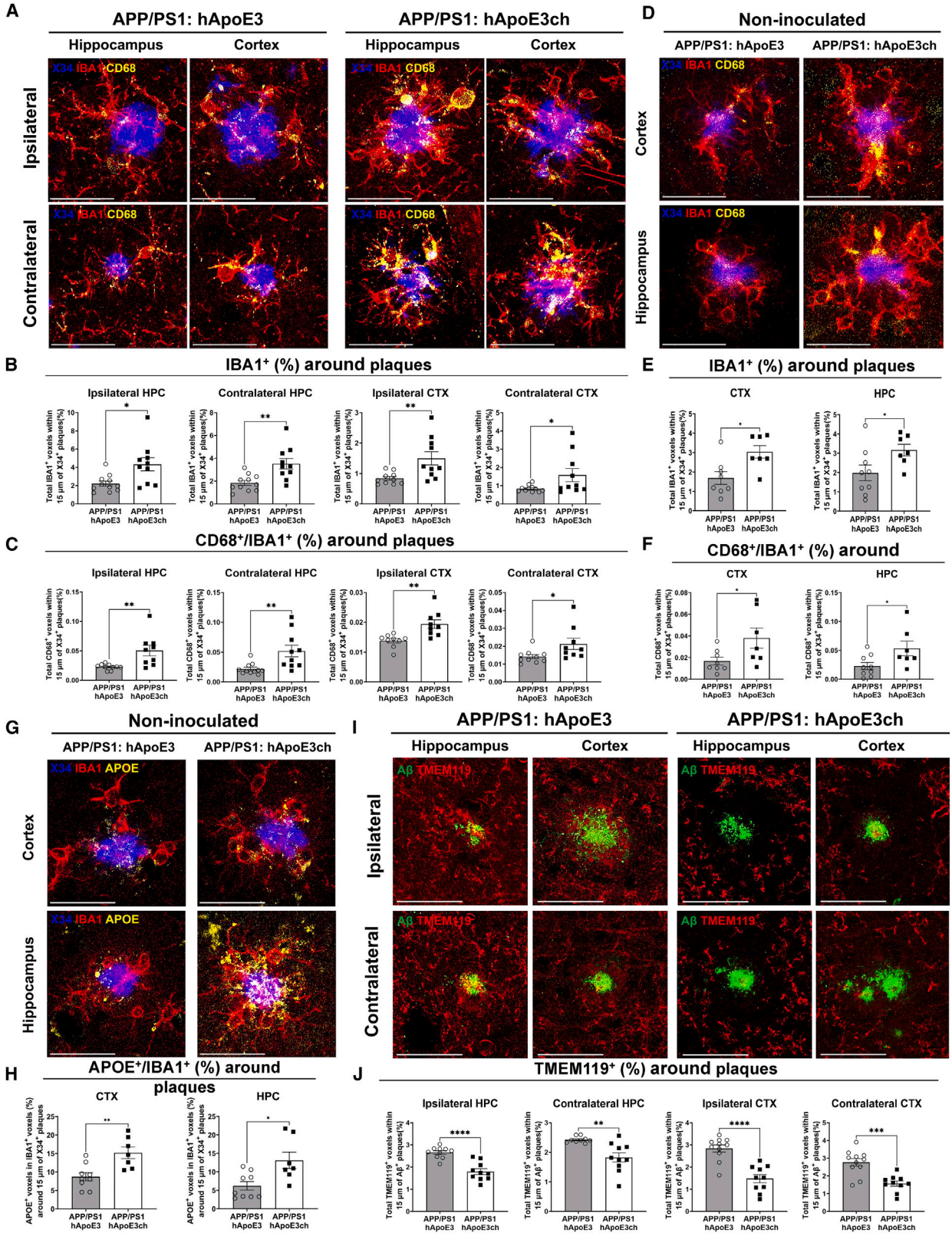
(A and B) Representative images of AT8 DAB staining of pTau (Ser202/Ser205) in AD-tau-seeded APP/PS1:APOE3 and APPPS1:APOE3ch mice and quantification of the percent of AT8 coverage. Ipsi: ipsilateral; Contra: contralateral. HPC: hippocampus (DG dentate gyrus and CA1 included); ENT: entorhinal cortex. Scale bars in A, 250μm. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(C and D) Representative confocal images of AT8-labeled (green) NP-tau pathology around X34$^+$ (blue) plaques in the AD-tau-seeded APP/PS1:APOE3 and APP/PS1:APOE3ch mice and quantification of the percent of AT8$^+$ voxels within 15μm around X34$^+$ plaques. HPC: hippocampus; CTX, cortex. Scale bars in C, 30μm. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(E and F) Representative confocal images of BACE1-labeled (yellow) dystrophic neurites around X34$^+$ (blue) plaques in the AD-tau seeded APP/PS1:APOE3 and APP/PS1:APOE3ch mice and quantification of the percent of BACE1$^+$ voxels within 15μm around X34$^+$ plaques. HPC: hippocampus; CTX, cortex. Scale bars in E, 30μm. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(G and H) Representative confocal images of synapsin-labeled (green) synapses around X34$^+$ (blue) plaques in the AD-tau-seeded APP/PS1:APOE3 and APOE3ch mice and quantification of the count of synapsin$^+$ puncta per μm^3 within 15μm around X34$^+$ plaques. HPC: hippocampus; CTX, cortex. Scale bars in G, 30μm. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

In all graphs, the statistical analyses were performed with student t-tests for mean values from individual animals. The error bars represent SEM values. The results designated with ns were not significant, whereas results designated with * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$) were significant. See also Figure S4 and Table S1.



(legend on next page)

pH-sensitive dye, pHrodoAPC, to recombinant fibrillar Aβ[25] (Figure S7A) and human tau fibrils (Figure S7B) and assessed their phagocytosis by bone marrow-derived macrophages (BMDMs) (Figures 7A–7D) differentiated from bone marrow of APOE3 and APOE3ch KI mice to assess myeloid cells that have similar properties to microglia.[18,26] Surprisingly, we observed no difference between APOE3 and APOE3ch BMDMs in phagocytosing Aβ fibrils at different concentrations (Figures 7A and 7B). Instead, we observed significantly increased phagocytosis of human tau fibrils by APOE3ch compared to APOE3 BMDMs (Figures 7C and 7D). To further assess the degradation of phagocytosed materials, we utilized a pH-insensitive dye, Alexa Fluor 488, to label recombinant human tau fibrils (Figure S7C) and monitored the decay of AlexFluor-488 dye within BMDMs for 48 h following 1 h of phagocytosis (Figure 7E).[18] Consistent with the pHrodo-labeled human tau phagocytosis, APOE3ch BMDMs showed increased initial uptake of human tau fibrils labeled with Alexa Fluor 488 (Figure 7F). However, Alexa Fluor 488 conjugated dye materials were rapidly digested by APOE3ch BMDMs and gradually returned to a similar level compared to APOE3 BMDMs within 48 h (Figure 7F). In addition, we performed a surface binding assay at 4°C with Alexa Fluor 488 labeled human tau fibrils and observed a concentration-dependent increase of human tau fibril binding to the APOE3ch and APOE3 BMDM cell surface (Figure S7D), suggesting that increased binding may enhance human tau fibril uptake.

Previous studies found that APOE and tau can bind similar surface receptors, including LRP1 and HSPGs.[27–31] We hypothesized that increased hTau fibril uptake in APOE3ch BMDMs could arise from decreased competition between APOE3ch and tau for surface receptors compared to APOE3. We performed a competitive hTau uptake assay utilizing *in vitro* lipidated recombinant human APOE3 and APOE3ch (lrhAPOE and lrhAPOE3ch) (Figure S7E). While the baseline of hTau uptake was higher in the APOE3ch groups, for both APOE3 and APOE3ch BMDMs hTau fibril uptake was significantly reduced by adding lrhAPOE3 but not lrhAPOE3ch (Figure 7G). The competitive uptake between hTau fibrils and lrhAPOE was directly monitored using a dual-labeled uptake assay (Figure S7F), where uptake of lrhAPOE3-488 was not significantly reduced by hTau (Figure S7G) but hTau-pHrodoAPC uptake was significantly reduced in APOE3 or APOE3ch BMDMs by lrhAPOE3-488 (Figure S7H). We next assessed the potential involvement of HSPGs and LRP1 in tau fibril uptake in BMDMs. Knockout of EXT1 or EXT2, which impairs HSPG surface expression, reduced hTau uptake by BV2 cells, a microglial cell line (Figure S7I). Addition of lrhAPOE3 further reduced hTau uptake in BV2 cells to a level close to that of EXT1/2 KO (Figure S7I). Addition of heparin sulfate or RAP protein to block HSPG or lipoprotein receptors, respectively also significantly reduced hTau uptake by BMDMs (Figure 7H). Consistent with these findings, heparin-beads pulled-down less lrhAPOE3ch than lrhAPOE3 (Figures 7J and 7K). Thus, the changes in the molecular interaction between APOE3ch and its receptors indirectly influence hTau phagocytosis by myeloid cells.

Since the effects of AD-tau seeding and spreading were greater in the presence of amyloid plaques, we aimed to investigate the potential synergistic effect of Aβ on tau when myeloid cells are challenged by both Aβ and hTau. We observed that phagocytosis of human Tau fibrils could be further enhanced by Aβ co-treatment at very low concentrations, in which APOE3ch BMDMs always phagocytosed more human Tau fibrils than APOE3 BMDMs (Figure 7L). In turn, the phagocytosis of Aβ42 fibrils could be enhanced by pre-treatment with human Tau (Figure 7M), but intriguingly this only occurred in APOE3ch BMDMs.

Microglia may promote tau seeding and spreading via release of seeding-competent tau.[32] To assess whether APOE3ch – expressing myeloid cells exhibited differences in the processing of phagocytosed tau, we developed a cell culture assay to

Figure 5. Enhancement of microglial response around plaques by APOE3ch

(A) Representative confocal images of IBA1 (red), CD68 (yellow) co-stained with X34$^+$ (blue) plaques in the AD-tau-seeded APP/PS1:APOE3 and APP/PS1:APOE3ch mice. Scale bars, 30 μm.

(B) Quantification of the percent of IBA1$^+$ voxels within 15μm around X34$^+$ plaques in different regions. HPC: hippocampus; CTX, cortex. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(C) Quantification of the percent of co-localized CD68$^+$/IBA1$^+$ voxels within 15μm around X34$^+$ plaques in different regions. HPC: hippocampus; CTX, cortex. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(D) Representative confocal images of IBA1 (red), CD68 (yellow) co-stained with X34$^+$ (blue) plaques in the non-inoculated APP/PS1:APOE3 and APP/PS1:APOE3ch mice (6m). Scale bars, 30 μm.

(E) Quantification of the percent of IBA1$^+$ voxels within 15μm around X34$^+$ plaques in the non-inoculated APP/PS1:APOE3 and APP/PS1:APOE3ch mice (6m). HPC: hippocampus; CTX, cortex. N = 9, APP/PS1: APOE3; N = 7, APP/PS1: APOE3ch.

(F) Quantification of the percent of co-localized CD68$^+$/IBA1$^+$ voxels within 15μm around X34$^+$ plaques in the non-inoculated APP/PS1:APOE3 and APP/PS1:APOE3ch mice (6m). HPC: hippocampus; CTX, cortex. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch.

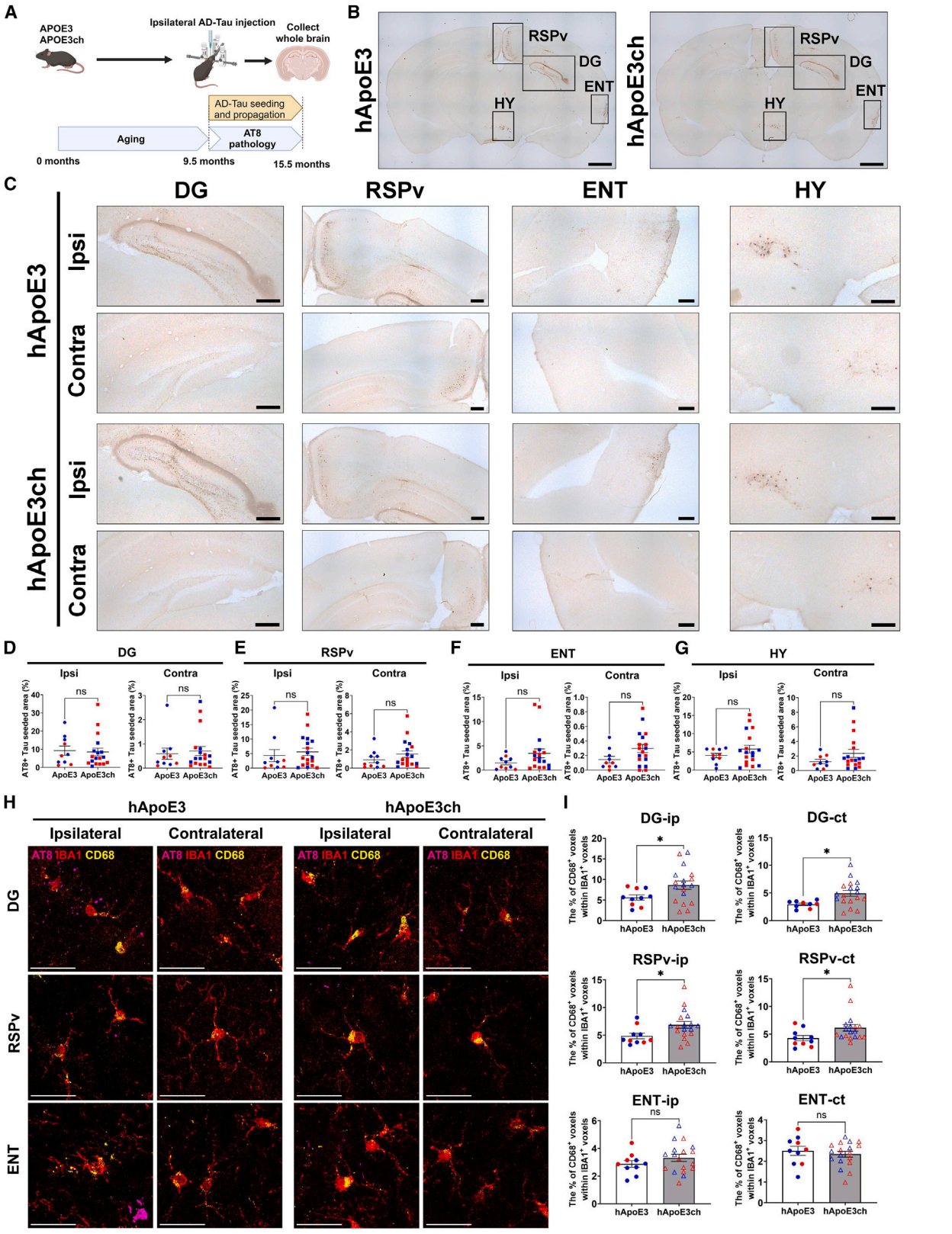
(G) Representative confocal images of IBA1 (red), APOE (yellow) co-stained with X34$^+$ (blue) plaques in the non-inoculated APP/PS1:APOE3 and APP/PS1:APOE3ch mice (6m). Scale bars, 30 μm.

(H) Quantification of the percent of APOE$^+$/IBA1$^+$ voxels within 15μm around X34$^+$ plaques in the non-inoculated APP/PS1:APOE3 and APP/PS1:APOE3ch mice (6m). HPC: hippocampus; CTX, cortex. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch.

(I) Representative confocal images of TMEM119 (red) co-stained with HJ3.4$^+$ (green) plaques in the AD-tau-seeded APP/PS1:APOE3 and APP/PS1:APOE3ch mice. Scale bars, 30μm.

(J) Quantifications of the percent of TMEM119$^+$ voxels within 15μm around Aβ$^+$ (HJ3.4-probed) plaques in different regions. HPC: hippocampus; CTX, cortex. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

In all graphs, the statistical analyses were performed with student t-tests for mean values from individual animals. The error bars represent SEM values. The results designated with ns were not significant, whereas results designated with * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$) were significant. See also Figures S2–S6 and Table S1.



(legend on next page)

monitor labeled hTau-fibril exocytotic release after phagocytosis (Figure S7J). Using negative gating based on non-Tau-488 media and regular APOE3/APOE3ch BMDM conditioned media and positive gating based on Tau-488 treatment media (Figures 7N and 7O), we monitored total Alexa Fluor 488$^+$ events in fresh media after a 24h challenge with the Tau-488 treatment media. Intriguingly, we observed a strong reduction of total Tau-488 release by APOE3ch BMDMs at different time points (Figure 7P), even though the residual Tau-488 within the APOE3ch group was initially significantly higher than the APOE3 group due to the enhanced uptake (Figure S7K). Overall, these data suggest that while APOE3ch microglia could degrade more phagocytosed hTau fibrils, they also release less hTau simultaneously and this may lead to less tau spreading *in vivo*. Additionally, the process of phagocytosis/degradation could be further stimulated by the presence of both amyloid and tau fibrils.

DISCUSSION

A recent report clearly demonstrated the remarkable resistance of the PSEN1-E280A carrier who was homozygous for the APOE3ch variant from amyloid-associated tau pathology and cognitive decline.[7,8] Both PET imaging and postmortem analysis indicated very high levels of Aβ pathology but very low levels of pTau pathology and PHFs in normally heavily affected brain regions. Whether APOE3ch directly affected amyloid pathology could not be answered by the data from the single human case report because of the lack of an age-matched APOE3 cohort. Notably, the original report predicted that the APOE3ch carrier would have developed even more Aβ pathology if the individual was homozygous for APOE3 based on their *in vitro* experiments assessing effects of APOE3ch and APOE3 on amyloid-β aggregation.[7] Supporting this prediction, we found a significantly reduced Aβ pathology in APP/PS1:APOE3ch at different ages with and without AD-tau seeding. The overall reactive microglial and astrocyte phenotypes were also attenuated in line with reduced amyloid pathology in APP/PS1:APOE3ch mice. This indicates a globally reduced reactive glial phenotype influenced by changes in Aβ plaque pathology. Second, an important and intriguing pathological observation from the case report was a disconnection between Aβ plaque burden and tauopathy in the temporal cortex.[7,8] While a definitive conclusion was not obtained as to whether APOE3ch resulted in the neuroprotection observed in the individual reported in the case report, here we found markedly reduced Aβ associated AT8$^+$ tau aggregates in the APP/PS1:APOE3ch vs. APP/PS1:APOE3 mice. Of note, the AT8$^+$ tau pathology around each X34$^+$ plaque was significantly reduced regardless of the total amyloid burden. Plaque-associated dystrophic neurites and synaptic loss were also reduced in the APP/PS1:APOE3ch mice. Whether APOE3ch protects against memory loss could not be assessed in the APP/PS1-21 mice, as we found no differences on several memory tests between APP/PS-1 mice and non-transgenic mice up to 6 months of age. Overall, these results suggest that in the presence of Aβ pathology, APOE3ch resulted in a local plaque-associated effect to decrease Aβ-associated tau seeding and spreading (Table S1).

Previous studies found that other AD genetic risk factors such as *TREM2*,[33] influence Aβ-associated tau seeding and altered plaque-associated microgliosis. We found that APOE3ch facilitates microglia clustering around plaques with enhanced phago-lysosomal reactivity, suggesting an elevated microglial activation state in the presence of Aβ plaques due to APOE3ch that could contribute to attenuated NP-tau pathology. Together, our results provide evidence of how APOE3ch improves microglial reactivity and prevents Aβ-induced NP-tau pathology similar to the previous case report. APOE3ch may reduce amyloid-dependent tau seeding and spreading at least in part through reduced competition for myeloid cell uptake and degradation of tau through shared receptors such as LRP1 or HSPGs. Whether this protection can be verified in APOE3ch heterozygotes or APOE2-like loss of LDL receptor binding ability variants remains to be elucidated. Whether protective effects of APOE3ch will be observed in late-onset AD and whether it will have different effects based on genetic ancestry are not clear. This will be important to assess as a recent study shows different risk for late-onset AD in APOE2 carriers based on genetic ancestry.[34] Another APOE3 protective variant, V236E, APOE3-Jacksonville (APOE3-Jac), reduced risk of developing late-onset AD, perhaps through a different mechanism.[35] The functional characterizations of APOE3-Jac revealed that V236E mutation protects against AD pathology by preventing self-aggregation of APOE and thus reducing the APOE-mediated Aβ deposition. On the contrary, due to a mutation in the C-terminal region of

Figure 6. Tau seeding and spreading in the non-amyloid APOE3 vs. APOE3ch mice

(A) Scheme of experimental timeline and pathological progress of the human AD-tau injection into B6 background non-amyloid APOE mice.

(B) Representative images of AT8 DAB staining of pTau (Ser202/Ser205) in AD-tau-seeded APOE3 and APOE3ch mice. Major tauopathy propagation regions are highlighted with black boxes. Scale bars, 1000 μm.

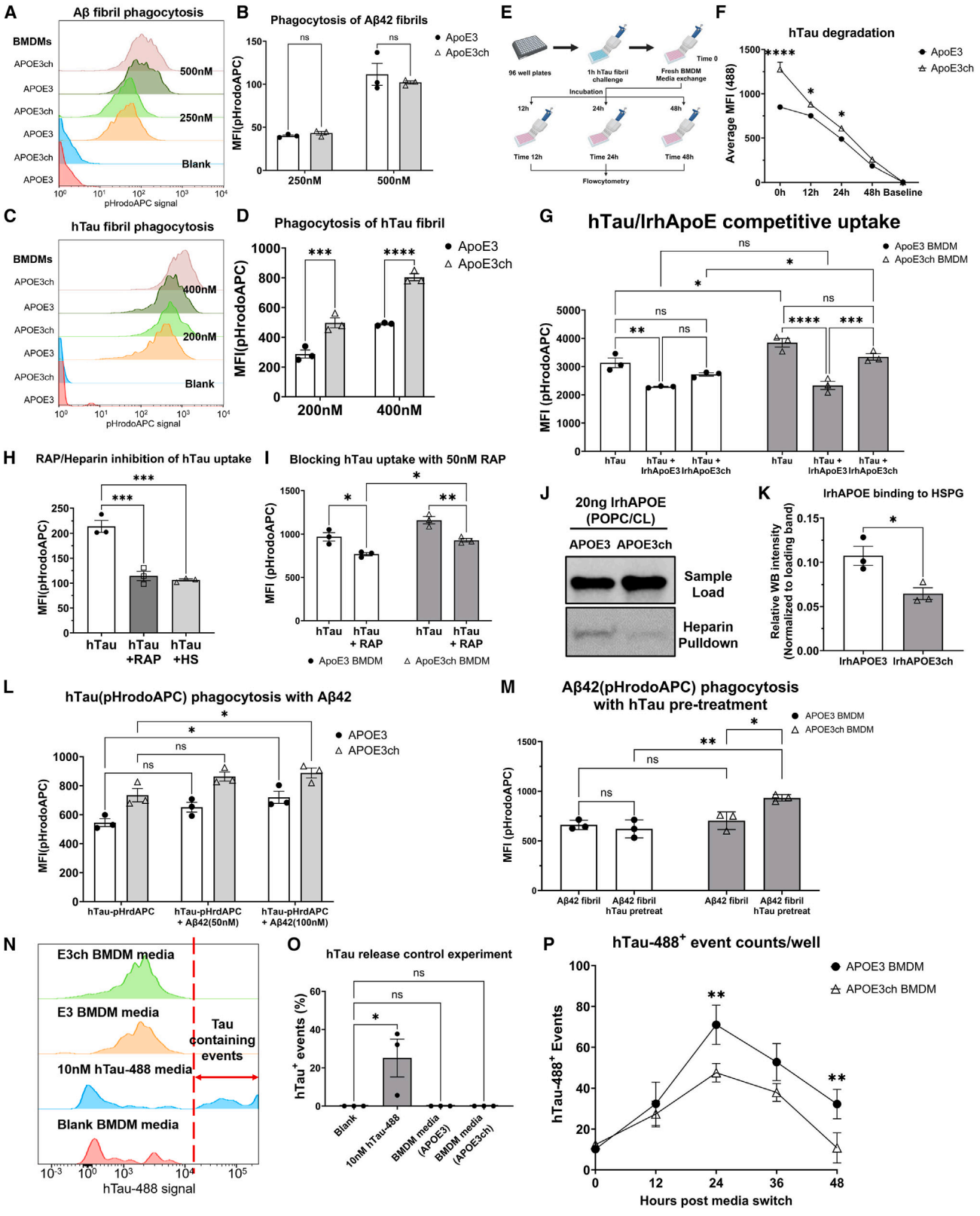
(C) Higher magnification images of the outlined regions highlighted in fig.6B. Scale bars, 250 μm, Ipsi: ipsilateral; Contra: contralateral. DG: dentate gyrus; RSPv: retrosplenial cortex; ENT: entorhinal cortex; HY: lateral hypothalamus.

(D–G) Quantification of total AT8 DAB-stained area (%) across both ipsilateral and contralateral (Ipsi and Contra) brain regions of AD-tau seeded B6 background non-pathogenic APOE mice. Ipsi: ipsilateral; Contra: contralateral. DG: dentate gyrus; RSPv: retrosplenial cortex; ENT: entorhinal cortex; HY: lateral hypothalamus. N = 6/4, 10m APOE3 males/females; N = 8/10, 10m APOE3ch males/females.

(H and I) Representative confocal images of AT8 (magenta), IBA1 (red), CD68 (yellow) in the representative pathological sites in AD-tau-seeded APOE3 and APOE3ch female mice and quantification of the percentage of CD68$^+$ voxels within IBA1$^+$ voxels. DG: dentate gyrus; ENT: entorhinal cortex; RSPv: retrosplenial cortex. Scale bars in H, 30μm. N = 6/4, 10m APOE3 males/females; N = 8/10, 10m APOE3ch males/females.

In all graphs, the statistical analyses were performed with student t-tests for averaged values from individual animals. The error bars represent SEM values. The results designated with ns were not significant, whereas results designated with * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$) were significant. Both sexes were combined into analysis after confirmation of no sex difference by two-way ANOVA. Males are colored blue and females are colored red.

See also Table S1.



(legend on next page)

APOE, APOE3-Jac also changes the stoichiometry between APOE protein and carrying lipids without affecting ligand-receptor interactions. This is different from APOE3ch which mostly impacts receptor binding ability including HSPG. The similarity and subtle differences at molecular level of two APOE3 protective variants, together with another recently discovered APOE4 R251G variant, could provide additional future insights into the molecular links between APOE and AD.

The R136S mutation was previously studied for its association with genetic lipid disorder HLP-III in APOE2 carriers.[36] At the same time, the first screening study to map the entire receptor binding site on APOE found that the APOE3 R136S variant resulted in 60% loss of receptor binding ability.[12] In our APOE3ch KI mouse model, we observed peripheral dyslipidemia similar to that seen in humans. Both male and female APOE3ch mice exhibited extremely high levels of plasma cholesterol, APOE and APOE-associated large lipoprotein particles such as vLDL and LDL. These are consistent with the idea that APOE3ch has decreased LDL receptor binding and causes HLP-III, similar to the mechanism of APOE2-associated HLP-III.[10] However, we detected no difference in either total cholesterol or APOE levels in the CSF, suggesting that APOE3ch has no major effect on CNS lipoprotein metabolism in the CSF. This is consistent with the fact that APOE-containing lipoproteins in the brain are derived from the CNS,[37] and not the periphery, and that the CNS contains only HDL-sized lipoproteins. A factor explaining differences between APOE3ch levels in the CNS and the periphery is that, regardless of the decrease in LDL receptor binding, there are different turnover rates of CNS APOE-associated lipoprotein production and recycling compared to the periphery.[38] However, we did detect some subtle changes in APOE and the expression of other proteins associated with APOE regulation or receptors such as ABCA1 and LRP1 that are sex-dependent. Whether and how APOE3ch affects homeostatic lipid metabolism and potentially influences CNS lipid metabolism and glial activation requires further study. Whether the sex differences of the protein levels among those lipid-associated proteins in the CNS would lead to phenotypic outcomes either in physiological functions or AD-related pathology also requires more studies.

Our *in vitro* studies show that APOE3ch myeloid cells have enhanced phagocytosis and degradation of Tau fibrils, which could be further induced with low concentrations of Aβ42. Another study has also found that certain toxic effects on neuronal function are worse when AD extracts contain both Aβ and tau.[39] Given that we did not observe differences in tau pathology in the absence of amyloidosis *in vivo*, the improvement in clearance of human tau might require priming of microglial responses beyond lysosomal reactivity that is seen in the APP/PS1:APOE3ch mice due to the presence of Aβ plaques. This idea is additionally supported by the similarly higher activation state of microglia in the 6-month-old APP/PS1:APOE3ch mice. Perhaps the effect of APOE3ch in microglial reactivity seen in response to amyloid is influencing specific microglial features such as phagocytosis and lysosomal degradation that influences NP-tau pathology but not all features or pathways linked to the DAM signature (Table S1).[40,41] We also showed that APOE3ch myeloid cells secreted less hTau material after phagocytosis, which could lend insight into the protective effect of APOE3ch in tauopathy propagation. Additionally, we provided evidence that the uptake of tau fibrils could be completed by lipidated APOE3 particles, but not APOE3ch particles. We verified potential tau receptors,[28,31] including HSPG and LRP1 on myeloid cells responsible for these effects. These observations would

Figure 7. Increased tau fibril uptake and degradation by APOE3ch BMDMs *in vitro*

(A) Representative images of overlay histogram demonstrating the shift in fluorescence of cells with pHrodo-APC labeled Aβ42 fibrils due to phagocytosis at 500nM concentration, 250nM and 0nM for both APOE3 and APOE3ch BMDMs.

(B) The averaged mean fluorescence intensity (MFI) +/− SEM (pHrodo-APC$^+$) of live BMDMs from each biological replicate corresponding to Figure 7A.

(C) Representative images of overlay histogram showing the shift in fluorescence of cells with pHrodo-APC labeled hTau (P301S) fibrils due to phagocytosis at 400nM concentration, 200nM and 0nM for both APOE3 and APOE3ch BMDMs.

(D) The averaged mean fluorescence intensity (MFI) +/− SEM (pHrodo-APC$^+$) of live BMDMs corresponding from each biological replicate to Figure 7C.

(E) Scheme of *in vitro* determination of hTau degradation following BMDM phagocytosis.

(F) The averaged mean fluorescence intensity (MFI, Alexa Fluor 488$^+$) of live BMDMs from each biological replicate corresponding to Figure 7E.

(G) The mean fluorescence intensity (MFI) +/− SEM (pHrodo-APC$^+$) of live BMDMs from each biological replicate with different hTau or hTau and lrhAPOE treatments as indicated.

(H) The mean fluorescence intensity (MFI) +/− SEM (pHrodo-APC$^+$) of live BMDMs from each biological replicate with different hTau treatments as indicated.

(I) The mean fluorescence intensity (MFI) +/− SEM (pHrodo-APC$^+$) of live BMDMs with hTau and RAP protein treatments as indicated.

(J and K) Western blot image of APOE from lipidated APOE with heparin-beads pull-down and quantification normalized to loading control with three independent experiments, K is mean ± SEM.

(L) The mean fluorescence intensity (MFI) +/− SEM (pHrodo-APC$^+$) of live BMDMs from each biological replicate with different concentrations of Aβ42 fibrils co-treated with 200nM hTau-pHrodo-APC fibrils.

(M) The mean fluorescence intensity (MFI) +/− SEM (pHrodo-APC$^+$) of live BMDMs from each biological replicate with different concentrations of Aβ42-pHrodo-APC fibrils pre-treated with 10nM hTau fibrils.

(N) Flow cytometry plot of overlay histogram showing the gating range of true hTau-488$^+$ events in the hTau exocytosis assay for Figures 7O and 7P.

(O) The mean percentage +/− SEM of true hTau-488$^+$ events in different samples as indicated.

(P) The counts of true hTau-488$^+$ events, mean ± SEM, in the BMDM media samples collected at different time points as indicated.

In all BMDM culture experiments, each data point represents the averaged MFI from technical triplicates of one biological replicate BMDM derived from labeled genotype (details in the methods). In all graphs, the statistical analyses were performed with two-way ANOVA and Šídák's multiple comparisons test for the adjusted p values. The error bars represent SEM values. The results designated with ns were not significant, whereas results designated with * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$) were significant according to adjusted p values in each indicated comparison.

See also Figure S7 and Table S1.

be important for potential therapeutic development focusing on the key role of amyloid-induced tau pathology for AD.

Limitations of the study

In this study, we utilized an AD-tau injection model to mimic the NP-tau seeding and spreading pathology in the AD patients. Although it is a useful mouse model for studying NP-tau pathology, this model has its own limitations: 1, it requires brain surgery; 2, the injected material contains more than human tau seeds; 3, there is no detectable neurodegeneration (i.e., loss of neurons). Additionally, we do not know the potential effects of the peripheral dyslipidemia caused by APOE3ch and the protective effect against CNS pathology. Finally, our mechanistic studies focused on myeloid cells (microglia and BMDMs). However, given that the expression of hTau receptors are not limited to microglia, it remains unclear whether the other CNS cells are involved in the protection of APOE3ch against ADAD.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- RESOURCE AVAILABILITY
  - Lead contact
  - Materials availability
  - Data and code availability
- EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS
  - Animals
  - Primary cells and cell lines
- METHOD DETAILS
  - General design of behavioral tests
  - Spontaneous alternation Y-maze
  - Conditioned fear
  - Brain and CSF collection
  - Plasma collection and cholesterol measurement
  - APOE enzyme-linked immunosorbent assay (ELISA)
  - Protein extraction and immunoblotting
  - Lipoprotein profiling
  - Negative stain transmission electron microscope (NS-TEM) imaging
  - AD-tau extraction and injection
  - DAB staining and quantification
  - Fluorescence staining and quantification
  - Confocal image acquisition and analysis
  - Aβ and hTau labeling and fibrillization
  - Thioflavin T (ThioT) Aβ fibrillization assay
  - APOE lipidation with POPC and cholesterol
  - Phagocytosis and degradation assay by flow cytometry
  - Heparin beads pull down assay
  - Labeled hTau fibril release measurement by flow cytometry
- QUANTIFICATION AND STATISTICAL ANALYSIS
  - Statistical analysis

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.cell.2023.11.029.

ACKNOWLEDGMENTS

This study was supported by the JPB Foundation (D.M.H.); Cure Alzheimer's Fund (D.M.H.); National Institutes of Health (NIH) grants RF1AG047644 (D.M.H.), RF1NS090934 (D.M.H.), Alzheimer's Association AARF-21-850865 (S.P.). The APOE3ch mice with human APOE3 R136S mutation were generated by Taconic Biosciences, Inc with the support of Cure Alzheimer's Fund and Rudolph Tanzi at Massachusetts General Hospital. The NS-TEM samples were prepared by WUCCI at Washington University in St. Louis, School of Medicine. The EXT KO BV2 cell line was a kind gift from Brestoff lab and GEiC center at Washington University in St. Louis, School of Medicine.

AUTHOR CONTRIBUTIONS

Y.C., J.D.U., and D.M.H. conceived and designed the study. Y.C. performed most of the experiments and analyzed the data, assisted by S.S., J.L., Y.Z., K.L., S.S., S.P., Y.Z., A.A.D., G.T.T., C.W., and J.P. S.P. collected mouse CSF and helped with image analysis and writing. M.R.S. prepared AD-tau stocks and performed partial AD-tau injections. H.J. generated in-house antibodies. C.M.Y. performed behavioral tests, helped with data analysis and writing. M.C. mentored and guided data interpretation and experimental designs. D.M.H. and J.D.U. supervised the research. Y.C., J.D.U., and D.M.H. wrote the manuscript with comments from the other authors.

DECLARATION OF INTERESTS

D.M.H. is an inventor on a patent licensed by Washington University to C2N Diagnostics on the therapeutic use of anti-tau antibodies. D.M.H. co-founded and is on the scientific advisory board of C2N Diagnostics. D.M.H. is on the scientific advisory board of Denali, Genentech, and Cajal Neuroscience and consults for Asteroid, and is on the Advisory Board for *Cell*.

M.C. is a member of Vigil Neuro scientific advisory board (SAB), is consultant for Cell Signaling Technology and NGM Bio. The rest of the authors have no conflict of interests.

Received: July 16, 2023
Revised: October 24, 2023
Accepted: November 27, 2023
Published: December 11, 2023

REFERENCES

1. Harper, L.C. (2022). Alzheimer's Association Facts and Figures. https://www.alz.org/alzheimers-dementia/facts-figures.
2. Long, J.M., and Holtzman, D.M. (2019). Alzheimer Disease: An Update on Pathobiology and Treatment Strategies. Cell *179*, 312–339.
3. Self, W.K., and Holtzman, D.M. (2023). Emerging diagnostics and therapeutics for Alzheimer disease. Nat. Med. *29*, 2187–2199.
4. Bateman, R.J., Aisen, P.S., De Strooper, B., Fox, N.C., Lemere, C.A., Ringman, J.M., Salloway, S., Sperling, R.A., Windisch, M., and Xiong, C. (2011). Autosomal-dominant Alzheimer's disease: a review and proposal for the prevention of Alzheimer's disease. Alzheimer's Res. Ther. *3*, 1.
5. Ryman, D.C., Acosta-Baena, N., Aisen, P.S., Bird, T., Danek, A., Fox, N.C., Goate, A., Frommelt, P., Ghetti, B., Langbaum, J.B.S., et al. (2014). Symptom onset in autosomal dominant Alzheimer disease: a systematic review and meta-analysis. Neurology *83*, 253–260.
6. Martens, Y.A., Zhao, N., Liu, C.-C., Kanekiyo, T., Yang, A.J., Goate, A.M., Holtzman, D.M., and Bu, G. (2022). ApoE Cascade Hypothesis in the pathogenesis of Alzheimer's disease and related dementias. Neuron *110*, 1304–1317.

7. Arboleda-Velasquez, J.F., Lopera, F., O'Hare, M., Delgado-Tirado, S., Marino, C., Chmielewska, N., Saez-Torres, K.L., Amarnani, D., Schultz, A.P., Sperling, R.A., et al. (2019). Resistance to autosomal dominant Alzheimer's disease in an APOE3 Christchurch homozygote: a case report. Nat. Med. *25*, 1680–1683.

8. Sepulveda-Falla, D., Sanchez, J.S., Almeida, M.C., Boassa, D., Acosta-Uribe, J., Vila-Castelar, C., Ramirez-Gomez, L., Baena, A., Aguillon, D., Villalba-Moreno, N.D., et al. (2022). Distinct tau neuropathology and cellular profiles of an APOE3 Christchurch homozygote protected against autosomal dominant Alzheimer's dementia. Acta Neuropathol. *144*, 589–601.

9. Wardell, M.R., Brennan, S.O., Janus, E.D., Fraser, R., and Carrell, R.W. (1987). Apolipoprotein E2-Christchurch (136 Arg— —Ser). New variant of human apolipoprotein E in a patient with type III hyperlipoproteinemia. J. Clin. Invest. *80*, 483–490.

10. Mahley, R.W., Huang, Y., and Rall, S.C., Jr. (1999). Pathogenesis of type III hyperlipoproteinemia (dysbetalipoproteinemia). Questions, quandaries, and paradoxes. J. Lipid Res. *40*, 1933–1949.

11. Mahley, R.W. (1988). Apolipoprotein E: cholesterol transport protein with expanding role in cell biology. Science *240*, 622–630.

12. Lalazar, A., Weisgraber, K.H., Rall, S.C., Jr., Giladi, H., Innerarity, T.L., Levanon, A.Z., Boyles, J.K., Amit, B., Gorecki, M., and Mahley, R.W. (1988). Site-specific mutagenesis of human apolipoprotein E. Receptor binding activity of variants with single amino acid substitutions. J. Biol. Chem. *263*, 3542–3545.

13. Mah, D., Zhu, Y., Su, G., Zhao, J., Canning, A., Gibson, J., Song, X., Stancanelli, E., Xu, Y., Zhang, F., et al. (2023). Apolipoprotein E Recognizes Alzheimer's Disease Associated 3-O Sulfation of Heparan Sulfate. Angew. Chem. Int. Ed. Engl. *62*, e202212636.

14. Radde, R., Bolmont, T., Kaeser, S.A., Coomaraswamy, J., Lindau, D., Stoltze, L., Calhoun, M.E., Jäggi, F., Wolburg, H., Gengler, S., et al. (2006). Abeta42-driven cerebral amyloidosis in transgenic mice reveals early and robust pathology. EMBO Rep. *7*, 940–946.

15. Palmisano, B.T., Zhu, L., Eckel, R.H., and Stafford, J.M. (2018). Sex differences in lipid and lipoprotein metabolism. Mol. Metab. *15*, 45–55.

16. He, Z., Guo, J.L., McBride, J.D., Narasimhan, S., Kim, H., Changolkar, L., Zhang, B., Gathagan, R.J., Yue, C., Dengler, C., et al. (2018). Amyloid-β plaques enhance Alzheimer's brain tau-seeded pathologies by facilitating neuritic plaque tau aggregation. Nat. Med. *24*, 29–38.

17. Leyns, C.E.G., Gratuze, M., Narasimhan, S., Jain, N., Koscal, L.J., Jiang, H., Manis, M., Colonna, M., Lee, V.M.Y., Ulrich, J.D., and Holtzman, D.M. (2019). TREM2 function impedes tau seeding in neuritic plaques. Nat. Neurosci. *22*, 1217–1222.

18. Gratuze, M., Chen, Y., Parhizkar, S., Jain, N., Strickland, M.R., Serrano, J.R., Colonna, M., Ulrich, J.D., and Holtzman, D.M. (2021). Activated microglia mitigate Aβ-associated tau seeding and spreading. J. Exp. Med. *218*, e20210542.

19. Horie, K., Salvadó, G., Barthélemy, N.R., Janelidze, S., Li, Y., He, Y., Saef, B., Chen, C.D., Jiang, H., Strandberg, O., et al. (2023). CSF MTBR-tau243 is a specific biomarker of tau tangle pathology in Alzheimer's disease. Nat. Med. *29*, 1954–1963.

20. Garai, K., Verghese, P.B., Baban, B., Holtzman, D.M., and Frieden, C. (2014). The binding of apolipoprotein E to oligomers and fibrils of amyloid-β alters the kinetics of amyloid aggregation. Biochemistry *53*, 6323–6331.

21. Wang, Y., Ulland, T.K., Ulrich, J.D., Song, W., Tzaferis, J.A., Hole, J.T., Yuan, P., Mahan, T.E., Shi, Y., Gilfillan, S., et al. (2016). TREM2-mediated early microglial response limits diffusion and toxicity of amyloid plaques. J. Exp. Med. *213*, 667–675.

22. Spires-Jones, T.L., and Hyman, B.T. (2014). The intersection of amyloid beta and tau at synapses in Alzheimer's disease. Neuron *82*, 756–771.

23. Jain, N., Lewis, C.A., Ulrich, J.D., and Holtzman, D.M. (2023). Chronic TREM2 activation exacerbates Aβ-associated tau seeding and spreading. J. Exp. Med. *220*, e20220654.

24. Ahmed, Z., Cooper, J., Murray, T.K., Garn, K., McNaughton, E., Clarke, H., Parhizkar, S., Ward, M.A., Cavallini, A., Jackson, S., et al. (2014). A novel in vivo model of tau propagation with rapid and progressive neurofibrillary tangle pathology: the pattern of spread is determined by connectivity, not proximity. Acta Neuropathol. *127*, 667–683.

25. Stine, W.B., Jungbauer, L., Yu, C., and LaDu, M.J. (2011). Preparing synthetic Aβ in different aggregation states. Methods Mol. Biol. *670*, 13–32.

26. Wang, S., Sudan, R., Peng, V., Zhou, Y., Du, S., Yuede, C.M., Lei, T., Hou, J., Cai, Z., Cella, M., et al. (2022). TREM2 drives microglia response to amyloid-β via SYK-dependent and -independent pathways. Cell *185*, 4153–4169.e19.

27. Kanekiyo, T., Zhang, J., Liu, Q., Liu, C.-C., Zhang, L., and Bu, G. (2011). Heparan sulphate proteoglycan and the low-density lipoprotein receptor-related protein 1 constitute major pathways for neuronal amyloid-beta uptake. J. Neurosci. *31*, 1644–1651.

28. Holmes, B.B., DeVos, S.L., Kfoury, N., Li, M., Jacks, R., Yanamandra, K., Ouidja, M.O., Brodsky, F.M., Marasa, J., Bagchi, D.P., et al. (2013). Heparan sulfate proteoglycans mediate internalization and propagation of specific proteopathic seeds. Proc. Natl. Acad. Sci. USA *110*, E3138–E3147.

29. Shinohara, M., Tachibana, M., Kanekiyo, T., and Bu, G. (2017). Role of LRP1 in the pathogenesis of Alzheimer's disease: evidence from clinical and preclinical studies. J. Lipid Res. *58*, 1267–1281.

30. Tachibana, M., Holm, M.-L., Liu, C.-C., Shinohara, M., Aikawa, T., Oue, H., Yamazaki, Y., Martens, Y.A., Murray, M.E., Sullivan, P.M., et al. (2019). APOE4-mediated amyloid-β pathology depends on its neuronal receptor LRP1. J. Clin. Invest. *129*, 1272–1277.

31. Rauch, J.N., Luna, G., Guzman, E., Audouard, M., Challis, C., Sibih, Y.E., Leshuk, C., Hernandez, I., Wegmann, S., Hyman, B.T., et al. (2020). LRP1 is a master regulator of tau uptake and spread. Nature *580*, 381–385.

32. Asai, H., Ikezu, S., Tsunoda, S., Medalla, M., Luebke, J., Haydar, T., Wolozin, B., Butovsky, O., Kügler, S., and Ikezu, T. (2015). Depletion of microglia and inhibition of exosome synthesis halt tau propagation. Nat. Neurosci. *18*, 1584–1593.

33. Colonna, M., and Wang, Y. (2016). TREM2 variants: new keys to decipher Alzheimer disease pathogenesis. Nat. Rev. Neurosci. *17*, 201–207.

34. Belloy, M.E., Andrews, S.J., Le Guen, Y., Cuccaro, M., Farrer, L.A., Napolioni, V., and Greicius, M.D. (2023). APOE Genotype and Alzheimer Disease Risk Across Age, Sex, and Population Ancestry. JAMA Neurol. e233599.

35. Liu, C.-C., Murray, M.E., Li, X., Zhao, N., Wang, N., Heckman, M.G., Shue, F., Martens, Y., Li, Y., Raulin, A.-C., et al. (2021). APOE3-Jacksonville (V236E) variant reduces self-aggregation and risk of dementia. Sci. Transl. Med. *13*, eabc9375.

36. Pocovi, M., Cenarro, A., Civeira, F., Myers, R.H., Casao, E., Esteban, M., and Ordovas, J.M. (1996). Incomplete dominance of type III hyperlipoproteinemia is associated with the rare apolipoprotein E2 (Arg136→ Ser) variant in multigenerational pedigree studies. Atherosclerosis *122*, 33–46.

37. Linton, M.F., Gish, R., Hubl, S.T., Bütler, E., Esquivel, C., Bry, W.I., Boyles, J.K., Wardell, M.R., and Young, S.G. (1991). Phenotypes of apolipoprotein B and apolipoprotein E after liver transplantation. J. Clin. Invest. *88*, 270–281.

38. Wildsmith, K.R., Basak, J.M., Patterson, B.W., Pyatkivskyy, Y., Kim, J., Yarasheski, K.E., Wang, J.X., Mawuenyega, K.G., Jiang, H., Parsadanian, M., et al. (2012). In vivo human apolipoprotein E isoform fractional turnover rates in the CNS. PLoS One *7*, e38013.

39. Ondrejcak, T., Klyubin, I., Hu, N.-W., O'Malley, T.T., Corbett, G.T., Winters, R., Perkinton, M.S., Billinton, A., Prenderville, J.A., Walsh, D.M., and Rowan, M.J. (2023). Tau and Amyloid β Protein in Patient-Derived Aqueous Brain Extracts Act Concomitantly to Disrupt Long-Term Potentiation in Vivo. J. Neurosci. *43*, 5870–5879.

40. Keren-Shaul, H., Spinrad, A., Weiner, A., Matcovitch-Natan, O., Dvir-Szternfeld, R., Ulland, T.K., David, E., Baruch, K., Lara-Astaiso, D., Toth,

B., et al. (2017). A Unique Microglia Type Associated with Restricting Development of Alzheimer's Disease. Cell *169*, 1276–1290.e17.

41. Chen, Y., and Colonna, M. (2021). Microglia in Alzheimer's disease at single-cell level. Are there common patterns in humans and mice? J. Exp. Med. *218*, e20202717.

42. Parhizkar, S., Gent, G., Chen, Y., Rensing, N., Gratuze, M., Strout, G., Sviben, S., Tycksen, E., Zhang, Q., Gilmore, P.E., et al. (2023). Sleep deprivation exacerbates microglial reactivity and Aβ deposition in a TREM2-dependent manner in mice. Sci. Transl. Med. *15*, eade6285.

43. Huynh, T.-P.V., Wang, C., Tran, A.C., Tabor, G.T., Mahan, T.E., Francis, C.M., Finn, M.B., Spellman, R., Manis, M., Tanzi, R.E., et al. (2019). Lack of hepatic apoE does not influence early Aβ deposition: observations from a new APOE knock-in model. Mol. Neurodegener. *14*, 37.

44. Wozniak, D.F., Diggs-Andrews, K.A., Conyers, S., Yuede, C.M., Dearborn, J.T., Brown, J.A., Tokuda, K., Izumi, Y., Zorumski, C.F., and Gutmann, D.H. (2013). Motivational disturbances and effects of L-dopa administration in neurofibromatosis-1 model mice. PLoS One *8*, e66024.

45. Chen, X., Firulyova, M., Manis, M., Herz, J., Smirnov, I., Aladyeva, E., Wang, C., Bao, X., Finn, M.B., Hu, H., et al. (2023). Microglia-mediated T cell infiltration drives neurodegeneration in tauopathy. Nature *615*, 668–677.

46. Yuede, C.M., Wozniak, D.F., Creeley, C.E., Taylor, G.T., Olney, J.W., and Farber, N.B. (2010). Behavioral consequences of NMDA antagonist-induced neuroapoptosis in the infant mouse brain. PLoS One *5*, e11374.

47. Garber, D.W., Kulkarni, K.R., and Anantharamaiah, G.M. (2000). A sensitive and convenient method for lipoprotein profile analysis of individual mouse plasma samples. J. Lipid Res. *41*, 1020–1026.

48. Guo, J.L., Narasimhan, S., Changolkar, L., He, Z., Stieber, A., Zhang, B., Gathagan, R.J., Iba, M., McBride, J.D., Trojanowski, J.Q., and Lee, V.M.Y. (2016). Unique pathological tau conformers from Alzheimer's brains transmit tau pathology in nontransgenic mice. J. Exp. Med. *213*, 2635–2654.

49. Garai, K., and Frieden, C. (2013). Quantitative analysis of the time course of Aβ oligomerization and subsequent growth steps using tetramethylrhodamine-labeled Aβ. Proc. Natl. Acad. Sci. USA *110*, 3321–3326.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Mouse monoclonal anti-human APOE	Generated in-house	HJ15.3
Mouse monoclonal anti-human APOE	Generated in-house	HJ15.7-biotinylated
Mouse monoclonal anti-human Aβ	Generated in-house	HJ3.4-biotinylated
Mouse monoclonal anti-mouse ABCA1	Generated in-house	HJ1
Mouse monoclonal anti-human MTBR-243 tau	Generated in-house	HJ32.11-biotinylated
Mouse monoclonal anti-phospho-tau (Ser202,Thr205) AT8-biotinylated	Invitrogen	Cat#MN-1020B; RRID: AB_223648
Rabbit monoclonal anti-APOB	Invitrogen	Cat#MA5-35458; RRID: AB_2849359
Goat polyclonal anti-APOJ	Invitrogen	Cat#PA5-46931; RRID:AB_2608081
Rabbit polyclonal anti-APOAI	Abcam	Cat#ab227455; RRID:AB_2892531
Rabbit polyclonal anti-LDLR	Invitrogen	Cat#PA5-90170; RRID:AB_2805946
Rabbit polyclonal anti-LRP1	Millipore Sigma	Cat#L2295; RRID:AB_10610711
Rabbit polyclonal anti-ACSL1	Invitrogen	Cat#PA5-78713; RRID:AB_2745829
Rabbit polyclonal anti-albumin	Cell Signaling Technology	Cat#4929; RRID:AB_2225785
Mouse monoclonal anti-GAPDH	Proteintech	Cat#60004-1; RRID:AB_2107436
Goat polyclonal anti-IBA1	Novus Biologicals	Cat#NB100-1028; RRID:AB_521594
Rabbit monoclonal anti-BACE1	Abcam	Cat#ab108394; RRID:AB_10861218
Mouse monoclonal anti-GFAP, Alexa Fluor 488–conjugated	BioLegend	Cat#644704; RRID:AB_2566109
Rat monoclonal anti-CD68	BioRad	Cat#MCA1957; RRID:AB_322219
Rabbit monoclonal anti-TMEM119	Cell Signaling Technology	Cat#90840; RRID:AB_2928137
Rat monoclonal anti-CLEC7A	InvivoGen	Cat#mabg-mdect; RRID:AB_2753143
Mouse monoclonal anti-Aβ 6E10, Alexa Fluor 647–conjugated	BioLegend	Cat#803021; RRID:AB_2783373
Biological samples		
AD-tau	Isolated from human AD brain (Guo et al.)	N/A
Chemicals, peptides, and recombinant proteins		
X-34	Millipore Sigma	Cat#SML1954; CAS: 215294-98-7
Human Aβ1-42 peptide	GenScript	Cat#RP10017; CAS: 107761-42-2
Human Recombinant Tau-441 (2N4R) P301S Mutant Protein Pre-formed Fibrils	StressMarq Bioscience Inc.	Cat#SPR-329
Human Recombinant APOE3 protein	AlexoTech	Cat#AE-101-10
Human Recombinant APOE3ch protein	AlexoTech; This paper	N/A
Recombinant RAP protein	Cell Sciences	Cat#CRM580B-His
Heparan sulfate sodium salt from bovine kidney	Millipore Sigma	Cat#H7640; CAS: 57459-72-0
Heparin Sepharose™ 6 Fast Flow Resin	Cytiva	Cat#17-0998-01
16:0-18:1 PC (POPC) 1-palmitoyl-2-oleoyl-*glycero*-3-phosphocholine	Avanti Polar Lipids	Cat#850457; CAS: 26853-31-6
Cholesterol (Plant)	Avanti Polar Lipids	Cat#700100; CAS: 57-88-5
pHrodo Deep Red TFP Ester	Invitrogen™	Cat#P35359
Alexa Fluor™ 488 NHS Ester (Succinimidyl Ester)	Invitrogen™	Cat#A20000
Thioflavin T	Laboratory of Albert Davis	N/A

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
3,3′,5,5′-Tetramethylbenzidine Liquid Substrate, Super Slow, for ELISA	Millipore Sigma	Cat#T5569
Pierce™ IP Lysis Buffer	Thermo Scientific™	Cat#PI87788
cOmplete™ Protease Inhibitor Cocktail	Millipore Sigma	Cat#4693116001
PhosSTOP™	Millipore Sigma	Cat#4906837001
Lumigen ECL Ultra	Lumigen	Cat#TMA-6
PierceTM ECL Western Blotting Substrate	Thermo Scientific™	Cat#32109
Elite ABC HRP Kit (Peroxidase Standard)	Vector Laboratories	Cat#PK-6100
Critical commercial assays		
Amplex™ Red Cholesterol Assay Kit	Invitrogen™	Cat#A12216
Pierce™ BCA Protein Assay Kit	Thermo Scientific™	Cat#23225
Experimental models: Cell lines		
Primary APOE3 bone marrow-derived macrophage	This paper	N/A
Primary APOE3ch bone marrow-derived macrophage	This paper	N/A
$BV2^{WT}$ microglial cell line	AcceGen	Cat#ABC-TC212S
$BV2^{EXT1-KO}$ microglial cell line	Laboratory of Jonathan R. Brestoff	N/A
$BV2^{EXT2-KO}$ microglial cell line	This paper	N/A
Experimental models: Organisms/strains		
Mouse: APOE3ch	This paper; Taconic Biosciences, Inc.	N/A
Mouse: APOE3	Huynh et al.	PMID: 31623648
Mouse: APP/PS1: APOE3ch	This paper; Laboratory of M. Jucker	PMID: 16906128
Mouse: APP/PS1: APOE3	Huynh et al.; Laboratory of M. Jucker	PMID: 31623648 PMID: 16906128
Software and algorithms		
ImageJ	Schneider et al.	https://ImageJ.nih.gov/ij/
Imaris 10 and XTensions	Oxford Instruments; Bitplane	https://imaris.oxinst.com/
Leica Application Suite X	Leica Microsystems	https://www.leica-microsystems.com/
NDP.view 2	Hamamatsu Photonics	https://www.hamamatsu.com/
FlowJo	FlowJo, LLC	https://www.flowjo.com/
Prism 9.0/10.0	GraphPad	https://www.graphpad.com/
Other		
JEM-1400Plus Transmission Electron Microscope	JEOL USA Inc.	https://www.jeol.com/
Nanozoomer 2.0-HT Slide Scanner	Hamamatsu Photonics	https://www.hamamatsu.com/
STELLARIS 5 Confocal Microscope	Leica Microsystems	https://www.leica-microsystems.com/
THUNDER Imaging System	Leica Microsystems	https://www.leica-microsystems.com/
BioTek Cytation 5 Cell Imaging Multimode Reader	Agilent	https://www.agilent.com/
Trans-Blot Turbo Transfer System	BioRad	https://www.bio-rad.com/
ChemiDoc Imaging System	BioRad	https://www.bio-rad.com/
BD FACSCalibur Flow Cytometer	BD Biosciences	https://www.bd.com/
BD FACSCanto II	BD Biosciences	https://www.bd.com/
BD LSRFortessa™ X-20 Cell Analyzer	BD Biosciences	https://www.bd.com/

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, David M. Holtzman (holtzman@wustl.edu).

Materials availability

The APOE3ch mouse line generated in this study will be available via the Cure Alzheimer's Fund. Further information and requests should be directed to the lead contact, David M. Holtzman (holtzman@wustl.edu).

The recombinant APOE3ch proteins were produced by AlexoTech, Inc. Product is available upon the query to mail@alexotech.com.

Data and code availability

(1) All data reported in this paper will be shared by the lead contact upon request.
(2) This paper does not report original code.
(3) Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

The APOE3ch homozygous mice were generated at Taconic Biosciences, Inc. as described.[43] For generation of APP/PS1: APOE3ch homozygous mice, 8-week-old APOE3ch mice were crossed to our APP/PS1: APOE3 mice to generate APP/PS1: APOE3ch E3/E3ch heterozygotes. Then the APP/PS1: APOE3chE3/E3ch heterozygotes were self-crossed between different littermates and then back crossed to APOE3ch homozygotes for APP/PS1: APOE3ch homozygotes. The mice used in this study were co-housed with APP/PS1: APOE3 that were at the same generation with gender match. At 6-month of age or 9.5-month of age, APP/PS1: APOE mice or APOE mice were injected with AD-tau extractions from the human brain as indicated in the RESULTS. Then the APOE3 and APOE3ch mice were sacrificed at age of 3 m and 10 m; the female APP/PS1: APOE mice were sacrificed at age of 6m; the AD-tau injected female APP/PS1: APOE mice were sacrificed at 3.5 months post injection (9.5m); the AD-tau injected APOE mice were sacrificed at 6 months post injection (15.5m). Genotypes and congenicity of final samples of >95% C57Bl background in this study were confirmed with tail samples by Transnetyx.

Primary cells and cell lines

We utilized primary bone marrow derived macrophages obtained from APOE3 and APOE3ch mice. We also utilized BV2 cell lines obtained from AcceGen and Brestoff lab at Washington University. These cells were cultured as described in the section below entitled "Phagocytosis and degradation assay by flow cytometry".

METHOD DETAILS

General design of behavioral tests

Male and female APP, E3Ch, and WT mice were evaluated for differences in behavior. Following habituation and handling in the Washington University Animal Behavior Core, mice were assessed on spontaneous alternation in a Y maze, and fear conditioning. All tests were conducted during the light phase of the light/dark cycle, by a female experimenter blind to the genotype of the mice.

Spontaneous alternation Y-maze

Testing was conducted according to our previously published procedures.[44,45] Briefly, this involved placing a mouse in one arm of a Y-maze that contained three arms that were 10.5 cm wide, 40 cm long and 20.5 cm deep where an arm was oriented at 120° with respect to each successive other arm. Mice were allowed to explore the maze for 8 min and entry into an arm was scored only when the hindlimbs had completely entered the arm. An alternation was defined as any three consecutive choices of three different arms without re-exploration of a previously visited arm. Dependent variables included the number of alternations and arm entries along with the percentage of alterations, which was determined by dividing the total number of alterations by the total number of entries minus 2, then multiplying by 100.

Conditioned fear

A conditioning protocol similar to previously described[46] was used to train and test mice using two clear-plastic conditioning chambers (26 × 18 × 18 cm high) (Med-Associates, St. Albans, VT) which were easily distinguished by different olfactory, visual, and tactile cues present in each chamber (Chamber 1 – metal grid floor, metal walls, mint scent; Chamber 2 – solid acrylic floor, black acrylic walls, coconut scent). On the first day, each mouse was placed into the conditioning chamber (Chamber 1) for 5 min and freezing behavior was quantified during a 2 min baseline period. Freezing (no movement except that associated with respiration) was quantified using FreezeFrame image analysis software (Actimetrics, Evanston, IL) which allows for simultaneous visualization of behavior while adjusting for a "freezing threshold" during 0.75 s intervals. After baseline measurements, a conditioned stimulus (CS) consisting of an 80 dB tone (white noise) was presented for 20 s followed by an unconditioned stimulus (US) consisting of a 1 s, 1.0 mA continuous foot shock. This tone-shock (T/S) pairing was repeated each minute over the next 2 min, and freezing was quantified after each of the three tone-shock pairings. Twenty-four hours after training, each mouse was placed back into the original conditioning

chamber (Chamber 1) to test for fear conditioning to the contextual cues in the chamber. This involved quantifying freezing over an 8 min period without the tone or shock being present. Forty-eight hours after the contextual test, the mice were evaluated on the auditory cue component of the conditioned fear procedure, which included placing each mouse into the other chamber (Chamber 2) containing distinctly different cues. Freezing was quantified during a 2 min "altered context" baseline period as well as over a subsequent 8 min period during which the auditory cue (CS) was presented.

Brain and CSF collection

Whole brain samples for AD-tau injected cohorts were collected after perfusion and fixed in the 4% paraformaldehyde (PFA) for 48h before immersion in 30% sucrose for cryopreservation for at least 48h. For non-AD-tau injected cohorts, the brains were dissected and cut into two hemispheres: the right hemisphere was fixed in 4% paraformaldehyde for at least 48 h and the left hemisphere was flash-frozen in 2-methylbutane on dry ice and subsequently stored at −80°C until needed. CSF was collected from animals from the cisterna magna through the dura matter by a glass capillary tube as previously described.[42,47] Samples with visual blood contamination were discarded after centrifugation.

Plasma collection and cholesterol measurement

In brief, 0.5 mL of blood from each mouse was collected from the cardiac artery by syringe before the perfusion. The plasma samples were separated from blood by centrifuge at 5000 rpm, 4°C for 10 min. Then 300 μL of each fresh plasma was aliquoted into 3x 100 μL vials. One aliquot was used on the same day as collection for lipoprotein separation by size-exclusion chromatography. The other two aliquots were flash-frozen by liquid nitrogen and kept at −80°C for further cholesterol measurements, ELISA or Western blot. The cholesterol measurements were performed with commercially available Amplex Red Cholesterol Assay Kit (Invitrogen, A12216) following the product protocol in duplicate.

APOE enzyme-linked immunosorbent assay (ELISA)

APOE ELISA was performed using in-house antibodies as previously described.[43] In brief, the high binding 96 well plates (half-area) were coated with 20 μg/mL anti-APOE antibody, HJ15.3, overnight at 4°C. The second day, the plates were washed in PBS 5x times and blocked with 2% BSA in the PBS. Then the samples were diluted in a sample buffer (0.5% BSA in PBS with protease inhibitors) and loaded into the plates in triplicates. The plates were incubated at 4°C overnight. On the third day, the plates were washed in PBS, followed by addition of the detection antibody, HJ15.7-biotinated, and incubation at 37°C for 90 min. The plates were then washed with PBS and incubated with streptavidin-poly-horseradish peroxidase-40 for 90 min at RT. Plates were then washed and developed with Super Slow ELISA 3,3′,5,5′-Tetramethylbenzidine (Sigma) and read at 650 nm.

Protein extraction and immunoblotting

Frozen hemispheres were homogenized and lysed with cold RIPA buffer in the presence of 1X Complete Protease Inhibitor and phosSTOP Phosphatase Inhibitor following the same procedure as described previously.[16] RIPA soluble fractions were subsequently aliquoted and flash-frozen in liquid nitrogen and stored at −80°C until needed. Upon Western blot assay, each aliquot was thawed on ice and the total protein concentration was determined by using BCA Protein Assay Kit. 10 μg of total protein mixed with 4x Laemmli sample buffer was loaded and separated by 4%–20% Mini-PROTEAN TGX gels in Tris/Glycine/SDS running buffer (Bio-Rad). For plasma WB, 30 μL from each frozen aliquot were mixed with 10 μL 4x Laemmli sample buffer, and then 20 μL mixture were loaded for each sample. For all WB gels, 5 vs. 5 samples were run on the same gel for representative images and the rest of samples were run on the second gel side by side to compare between groups. Gels were transferred on a PVDF membrane using the Trans-Blot Turbo Transfer System (Bio-Rad). The membrane was blocked with 5% milk in TBS with 0.05% Tween (TBST) for 60 min at room temperature. The following primary antibodies in this study included anti-APOB (1:1000; rabbit monoclonal, Invitrogen, MA5-35458), anti-APOE (1:2000; mouse monoclonal, generated in-house, HJ15.7), anti-APOJ (1:1000; goat polyclonal, Invitrogen, PA5-46931), anti-APOAI (1:1000; rabbit polyclonal, Abcam, ab227455), anti-LDLR (1:1000; rabbit polyclonal, Invitrogen, PA5-90170), anti-ABCA1(1:1000; mouse monoclonal, generated in-house, HJ1), anti-LRP1 (1:1000; rabbit polyclonal, Sigma, L2295), anti-ASCL1 (1:1000; rabbit polyclonal, Invitrogen, PA5-78713), anti-Albumin (1:4000; rabbit polyclonal, Cell Signaling Technology, 4929) and anti-GAPDH (1:5000; mouse monoclonal, Proteintech, 6004-1-1g). After primary antibody incubation in the blocking solution at 4°C overnight, membranes were washed and incubated in HRP conjugated secondary antibodies according to the species of primary antibodies for 1 h at room temperature.

Then membranes were washed again and developed using Lumigen ECL Ultra (TMA-6), then gels of males and females (total 4 gels) were imaged together by ChemiDoc Imaging Systems (Bio-Rad). The plasma anti-Albumin WB was particularly developed with Pierce ECL Western Blotting Substrate (Thermo Scientific, 32109) due to the abundance of the protein.

For sequential blots of multiple proteins, the primary and secondary antibodies were removed by incubating the blots in the stripping buffer (62.5mM Tris pH 6.8, 2% SDS, 0.7% beta-mercaptoethanol) at 55°C for 20 min with agitation. Stripping buffer was disposed, and blots were rinsed first with deionized water for 1–2 min, next with Tris-Buffered Saline with 0.125% Tween 20 (TBS-T) for 3 times (10 min each). The membranes were then used following the SDS-PAGE method above. All raw WB images were shown in the Figure S1.

Lipoprotein profiling

To profile lipoproteins in the plasma, 100 μL of fresh plasma samples were run through the Fast Protein Liquid Chromatography (FPLC) tandemed with size-exclusion chromatography column Superdex 200 Increase 10/300 GL (Cytiva inc.) and collected into 1mL fractions in PBS. The cholesterol concentration in each fraction was then determined by the same kit as described in METHOD to map the corresponding peak of vLDL, LDL and HDL lipoparticles based on standard curve matching with the previous literature[47] generated with commercially available pure vLDL, LDL and HDL. The profiling from each group was averaged from 5 biological replicates and individual plots were attached in the Figure S1. Intact lipoproteins were confirmed with negative staining using a JEOL JEM-1400Plus transmission electron microscope (JEOL USA Inc., Peabody, MA).

Negative stain transmission electron microscope (NS-TEM) imaging

All NS-TEM grids were stained and prepared by the WUCCI center. Prepared samples were imaged on a JEOL-1400Plus transmission electron microscope (JEOL USA Inc., Peabody, MA) operating at an accelerating voltage of 120 kV using a NanoSprint15-MkII 16-megapixel scientific complementary metal-oxide semiconductor (sCMOS) camera (Advanced Microscopy Techniques, Woburn, MA).

AD-tau extraction and injection

AD-tau was isolated from a human AD brain, Braak stage VI, as described previously.[48] In brief, the frontal cortical gray matter from human AD brain was homogenized using a Dounce homogenizer in nine volumes (v/w) of high-salt buffer (10 mM Tris-HCl, pH 7.4, 0.8 M NaCl, 1 mM EDTA, and 2 mM dithiothreitol (DTT), with protease inhibitor cocktail, phosphatase inhibitor, and PMSF) with 0.1% sarkosyl and 10% sucrose added and centrifuged at 10,000 g for 10 min at 4°C. Pellets were reextracted twice using the same buffer conditions, and the supernatants were filtered and pooled. Additional sarkosyl was added to reach 1%. After 1h rotation at room temperature, samples were centrifuged again at 300,000 g for 60 min at 4°C. The 1% sarkosyl-insoluble pellets, which contain pathological tau, were washed once in PBS and then resuspended in PBS (~100 μL/g gray matter) by passing through 27-G 0.5-in. needles. The resuspended sarkosyl-insoluble pellets were further purified by a brief sonication (20 pulses at ~0.5 s/pulse) followed by centrifugation at 100,000 g for 30 min at 4°C. The pellets were resuspended in PBS at one-fifth to one-half of the pre-centrifugation volume, sonicated with 20–60 short pulses (~0.5 s/pulse), and spun at 10,000 g for 30 min at 4°C to remove large debris. The final supernatants, which contained enriched AD PHFs, were used in the study and referred to as AD-tau.

Stock AD-tau aggregates were prepared as previously described.[20,49] The stock concentration of the preps used in this study was 2.6 μg/μL. The aggregates were further diluted to a final concentration of 0.4 μg/μL and aliquoted into 6 μL vials. Before the injection, one vial per mouse was thawed on ice and sonicated in a water bath sonicator (QSonica, Q700) for 30 s at 60% amplitude at 4°C. Mice receiving injections were anesthetized with isoflurane and immobilized in a sterotactic frame (David Kopf Instruments). 2 μg AD-tau (1 μg at each injection site) in the dentate gyrus (bregma: −2.5 mm; lateral: −2.0 mm; depth: −2.2 mm) and overlying cortex (bregma: −2.5 mm; lateral: −2.0 mm; depth: −1.0 mm) was ipsilaterally injected using a Hamilton syringe (Hamilton; syringe: 80265–1702RNR; needle: 7803–07). Mice were allowed to recover on a 37°C heating pad and monitored for the first 72 h after surgery.

DAB staining and quantification

Serial coronal sections (30 μm thickness) were collected from frontal cortex to caudal hippocampus (whole brain) using a freezing sliding microtome (ThermoFisher). Three sections from each animal were selected for AT8-biotinylated and two sections were selected for HJ3.4-biotinaylated immunohistochemical staining. Sections were washed in Tris-buffered saline (TBS) buffer for 3 × 5 min. After washing, sections were incubated in 0.3% hydrogen peroxide in TBS for 10 min at room temperature in order to block endogenous peroxidase activity, followed by 3 × 5 min TBS washes. Sections were then blocked with 3% milk in TBS with 0.25% Triton X-100 (TBSX) for 1 h at room temperature. Following this, sections were incubated at 4°C overnight with AT8-biotinylated antibody (mouse monoclonal, 1:500) or HJ3.4-biotinylated antibody (mouse monoclonal, 1:500) respectively. The next day, after 3 × 5 min washing in TBS, sections were incubated in ABC Elite solution (VectaStain, PK-6100) for 1 h, followed by 3 times washing in TBS. Finally, sections were developed and stained using DAB Peroxidase Substrate, washed, and mounted on slides. After drying, the slides were dehydrated in increasing ethanol concentrations followed by xylene, coverslipped with Cytoseal 60. The AT8 staining slices were scanned using Hamamatsu's Nanozoomer microscope at 20X magnification and the HJ3.4 staining slices were scanned by THUNDER imaging system (Leica) at 2.5X. Images were analyzed using the NDP.view 2. Images were further processed and quantified with the use of Fiji software.

Fluorescence staining and quantification

The immunofluorescence (IF) staining co-stains in this study were designed as (1) X34, BACE1, AT8, and 6E10; (2) X34, GFAP, Iba1, and ApoE; (3) X34, Iba1, and CD68; (4) HJ3.4, TMEM119, CLEC7A or (5) AT8, IBA1, CD68. Fibrillar Aβ was stained by X34 dye (Sigma, SML-1954) and antibodies to phosphorylated Tau (1:500; mouse monoclonal AT8, Invitrogen, MN1020B), IBA1 (1:500; goat polyclonal, Novus Biologicals, NB100-1028), BACE1 (1:500; rabbit monoclonal, Abcam, ab108394), GFAP (1:2000; mouse monoclonal, Alexa Fluor 488–conjugated, BioLegend, 644704), APOE (1:500; generated in-house, HJ15.7-biotinylated; 1:500), CD68 (1:500; rat monoclonal, Bio-Rad, MCA1957), TMEM119 (1:500; rabbit monoclonal, Cell Signaling Technology, 90840), CLEC7A (1:50 at room

temperature; rat monoclonal; InvivoGen, mabg-mdect), and β-amyloid (1:1000; mouse monoclonal 6E10, Alexa Fluor 647–conjugated, BioLegend, 803021) were used to evaluate peri-plaque pathologies. Fluorescent secondary antibodies were selected according to the species of primary antibodies and channels of interest, and all used at 1:1000 dilution.

For each animal, two slices at the same positions were selected. Rostral slices without the hippocampus were specifically selected for non-tau-seeded IHCs. Free-floating sections were washed three times and then permeabilized in 0.25% Triton X-100 PBS (PBS-X) for 30 min. Tissue sections were then incubated in 40% ethanol (EtOH)/60% PBS dissolved X34 for 20 min, washed in X34 buffer (40% EtOH in PBS), and washed in PBS twice. Sections were incubated in blocking solution for 30 min (3% BSA, 3% normal donkey serum, and 0.1% PBS-X) and then incubated in primary antibodies at indicated concentrations in blocking solution overnight at 4°C. The next day, sections were washed three times, placed in secondary antibodies for 2 h at room temperature, and then washed again three times for 20 min. Lipofuscin was quenched with 0.1% Sudan black and washed once in 0.05% PBS–Tween 20 and again in PBS. Sections were mounted, sealed in ProLong Gold anti-fade (Thermo Fisher Scientific, P36930), and stored in the dark at 4°C until imaging.

Confocal image acquisition and analysis

Images were obtained with 8 replicates from two slices per brain region per mouse for IF. In brief, four z-stacks (20 μm) of each brain region (e.g., four from ipsilateral CTX and four from contralateral CTX) per slice were acquired on a Leica STELLARIS 5 confocal microscope and the Leica Application Suite X software (4.2.1.23810) with a 20× objective and 1,024× 1,024 resolution. Laser and detector settings were set for the acquisition of each immunostaining combination consistently. High-resolution representative images were maximum projection from stacks imaged by confocal microscopy were 3D-reconstructed using the Leica Application Suite X software. Quantification of confocal images for 6E10, AT8, BACE1, IBA1, CD68, TMEM119, CLEC7A and APOE around $X34^+$ plaques was performed on a semiautomated platform using MATLAB and Imaris 10 software (Bitplane) to create surfaces of each stain based on a threshold applied to all images, dilate X34 surfaces 15 μm, and colocalize various immunostained surfaces and dilated X34 surfaces. For percent voxel quantifications, colocalized volumes for each immunostained surface were summed per image and divided by the $X34^+$ extended surface. Quantification of confocal images for CD68 and APOE in $Iba1^+$ microglia or $GFAP^+$ astrocytes was performed by creating a surface of CD68/Iba1 costaining and dividing it with total IBA1 or GFAP surfaces. The final value was averaged from a total of 8 images per brain region per mouse. All staining experiments were imaged and quantified by a blinded investigator or labeling strategy.

Aβ and hTau labeling and fibrillization

Human Aβ42 peptide was purchased from GenScript (RP10017). Monomeric Aβ42 was prepared in dimethyl sulfoxide following previous literature and conjugated with pHrodo Deep Red TFP Ester (succinimidyl ester) (Invitrogen, P35359) (pHrodoAPC) at molar ratio protein:dye = 1:5 in sodium bicarbonate buffer (pH 8.5) at room temperature overnight. Unconjugated dye was removed after 24-h 3.5-kDa cutoff dialysis in H2O at 4°C. Then, Aβ42-pHrodoAPC peptide was dried with speed vacuum at room temperature overnight. Aβ42-pHrodoAPC fibrils were prepared following the same protocol20 and dialyzed to phosphate-buffered saline (PBS) for 3 days. Fibril samples were confirmed with negative staining using a JEOL JEM-1400Plus transmission electron microscope (JEOL USA Inc., Peabody, MA). Final products were prepared into 10 μM aliquots and stored at −80°C until further use. Human recombinant Tau-441 (2N4R) P301S mutant protein pre-formed fibrils were purchased from StressMarq Bioscience Inc. (SPR-329) and labeled with either pHrodo Deep Red TFP Ester (succinimidyl ester) (Invitrogen, P35359) (pHrodoAPC) or Alexa Fluor 488 NHS Ester (Invitrogen, A20000) in the same procedure as described. Final hTau-pHrodoAPC and hTau-488 fibril products were prepared into 10 μM aliquots and stored at −80°C until needed.

Thioflavin T (ThioT) Aβ fibrillization assay

The detailed protocol was originally developed in previous studies.[20,49] In this study, 0.5 mg of Aβ1–42 peptide (GenScript, RP10017) was dissolved in 4 M GdnCl and PBS at a 1mM stock solution. 1 mM Aβ42 peptide was diluted with PBS to final concentration at 20 μM with and without 2 μM recombinant APOE3 or APOE3ch, and with 20 μM ThioT. The final concentration of GdnCl was adjusted to 80 mM in all samples. Then each sample was transferred to a clear-bottom 96 well plate in triplicate and fluorescence (λex = 438 nm, λem = 480 nm) was recorded for 20 h with 1 min shaking and 5 min reading intervals using a BioTek Cytation 5 Cell Imaging Multimode Reader. Each symbol represents the average value from triplicate samples at the indicated time point.

APOE lipidation with POPC and cholesterol

Recombinant human APOE3 and APOE3ch proteins were purchased from AlexoTech (AlexoTech, AE-101-10; APOE3ch requires additional query). To produce lipidated APOE particles, 18.62 μL of POPC (10 mg/mL in chloroform) and 1.89 μL of Cholesterol (10 mg/mL in chloroform) were mixed and dried in nitrogen. Then 50 μL of DPBS were added to resuspend lipids at RT for 30 min after vortex, followed by adding 16.56 μL Na-cholate (45 mg/mL). After vortex and sitting at room temperature for 30 min–1 h, the mixture should become clear. If not, 1/3 volume of Na-cholate (45 mg/mL) was added again. Once lipids were completely dissolved, 167 μg of APOE was added and the mixture solution is then to sit at room temperature for 45 min–1 h. Finally, after PBS dialysis at 4°C for 72 h, the prep was stored at 4°C for no more than 2 weeks prior to experiments.

Phagocytosis and degradation assay by flow cytometry

Bone marrow from individual APOE3 and APOE3ch mice was isolated from femur after perfusion and frozen in one aliquot per mouse in liquid nitrogen. Before the phagocytosis or degradation assay, three vials of bone marrow of 3 different animals from each genotype were differentiated in BMDM differentiation media (RPMI 1640 medium, 10% FBS, 10% L929 supernatant, 1× Pen-Strep, 1× Glutamax, 1× sodium pyruvate, and 1× nonessential amino acids) for 4–5 days. When fully differentiated BMDMs were 70% confluent, non–tissue culture–treated 96-well plates were seeded with 20,000 cells per well in triplicates for each biological replicate in BMDM media for a phagocytosis/degradation assay the next day.

For the phagocytosis assay, Aβ42-pHrodoAPC or hTau-pHrodoAPC fibrils were added into each well at final concentration as indicated in RESULTS in RPMI 1640 serum free media for 1 h. For degradation assay, all cells were treated with hTau-488 fibrils for 1 h and switched to normal media for 12, 24, and 48 h before being harvested. For surface binding assay, pre-chilled BMDMs were treated with hTau-488 fibrils for 1 h with concentrations indicated in RESULTS at 4°C. For competitive phagocytosis assay, additional reagents such as recombinant RAP protein (Cell Sciences, CRM580B-His), heparan sulfate sodium salt (Sigma, H7640-1MG) and lipidated APOE particles as indicated in RESULTS were added with hTau-pHrodoAPC fibrils together in serum free media and then added into each well. After washing with PBS twice, BMDMs were resuspended with MACS buffer (PBS, 1 mM EDTA, and 0.5% BSA), and flow cytometry was performed on a BD FACS Calibur/BD X20/BD Canto II System at the flow cytometry core at Washington University in St. Louis. Alexa Fluo 488 signal was determined by FL1 (FITC) and Deep Red pHrodo signal was determined by FL4 Far Red (APC) after pre-gating with FL3 Red Propidium Iodide staining (PE) and further analyzed by FlowJo. BV2 phagocytosis assay with hTau fibrils was performed following the same protocol with experimental replicates.

Heparin beads pull down assay

Heparin beads resin was purchased from (Cytiva, 17099801). In brief, 100 ng of recombinant hAPOE3, hAPOE3ch and freshly prepared lipidated rhAPOE3/rhAPOE3ch were used for heparin beads pulldown. First, 10 μL of 50% heparin bead slurry (in 20% ethanol) was equilibrated by PBS buffer with additional150nM NaCl. Then 100 ng of each protein were mixed with 10 μL slurry and additional PBS buffer with additional 150nM NaCl to bring up the volume to 1 mL. Each vial was rotated at room temperature for 1 h, then spun down at 1000g for 5 min. The beads were collected after discarding supernatants and the proteins captured were eluted by 50 μL of 1x SDS-loading dye in PBS buffer. Together with another 100 ng of each protein sample, all protein samples were used to run regular APOE WB.

Labeled hTau fibril release measurement by flow cytometry

As described above, hTau-488 fibrils were premade and stored with 10 μM aliquots. BMDMs were treated with 10 nM hTau-488 fibrils in serum free media for 24 h at 37°C. After 24 h, BMDMs were washed with PBS twice and then cultured back with normal BMDM media. At each time point, 100 μL of the media were collected and run the flow cytometry on a BD FACS Canto II System at the flow cytometry core at Washington University in St. Louis. The control samples were prepared with new media, 10 nM hTau-488 media and regular APOE3/APOE3ch media collected from additional BMDM culture. In the flow cytometry, Alexa Fluo 488 signal was determined by FITC channel after pre-gating with Red Propidium Iodide staining and further analyzed by FlowJo.

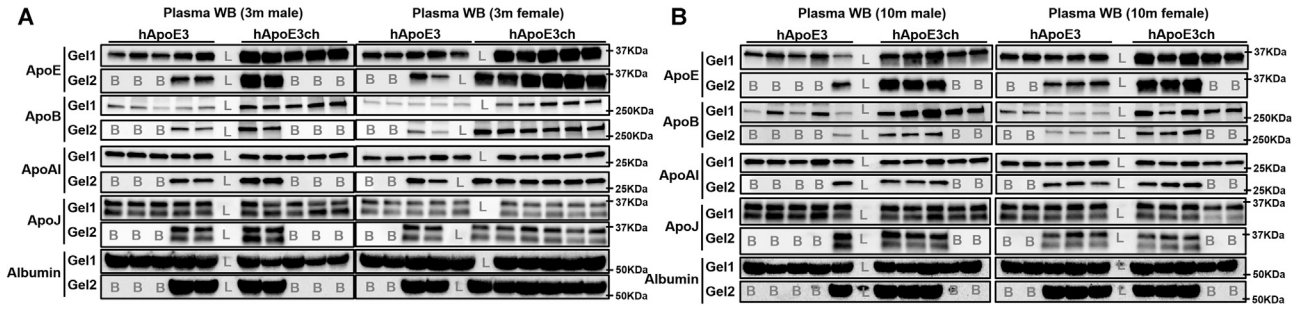
QUANTIFICATION AND STATISTICAL ANALYSIS

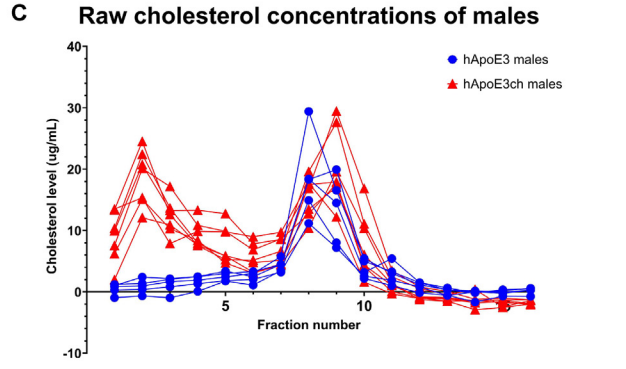
Statistical analysis

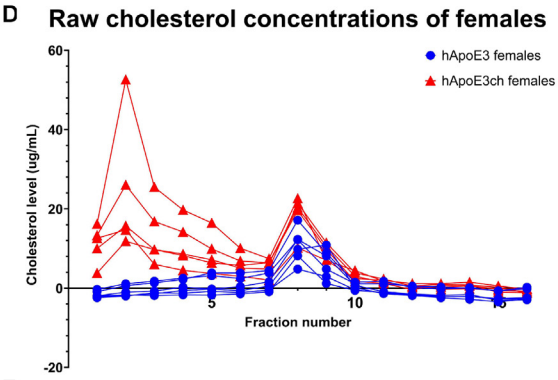
GraphPad Prism 9.0 was used to perform all statistical analyses. Data are presented as means ± SEM. Data were checked for normality using the Shapiro-Wilk method, D'Agostino and Pearson normality test, and Kolmogorov-Smirnov normality test. The statistical outliers were excluded by the ROUT method (Q = 1%) in GraphPad Prism. Statistical significance between groups with normally distributed data was calculated by Student's *t* test or two-way ANOVA followed by Šídák's multiple comparisons test or Tukey's post hoc test for group-wise comparisons. $p < 0.05$ was considered significant: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$). More details can be found in the figure legends.

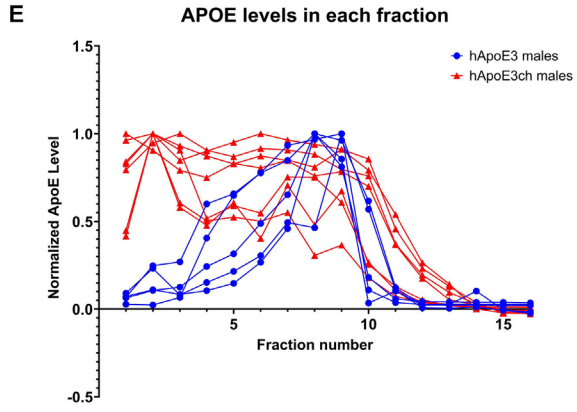
For animal behaviors, all statistical analyses were performed using GraphPad Prism (v.10). Analysis of variance (ANOVA), (one-way, factorial, repeated measures) were used to analyze the behavioral data. With a statistically significant interaction between main factors, simple main effects were calculated to provide clarification of statistically significant between-genotype and within-genotype differences. Bonferroni correction was applied to multiple pairwise comparisons and post hoc tests. Probability value for all analyses was $p < 0.05$, unless otherwise stated. More details can be found in the legend to Figure S2.

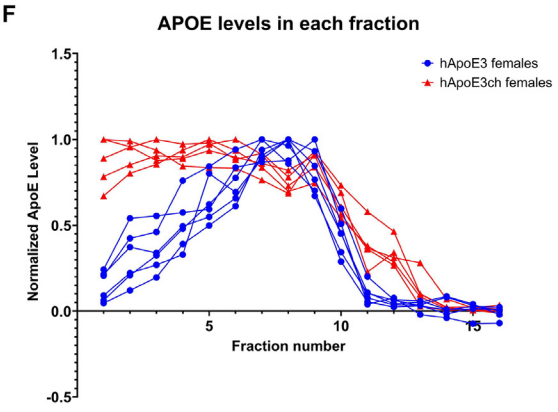
Supplemental figures

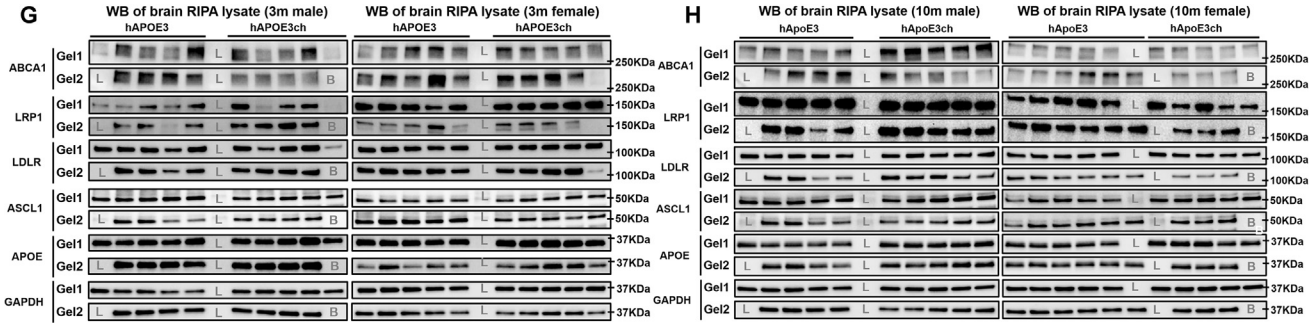












(legend on next page)

Figure S1. Individual plots of plasma lipoprotein curves and WB of 3 months (m) and 10 m cohorts, related to Figures 1€ and 2

(A and B) Raw Western blot images used for quantification results shown in Figures 1I–1L and 1M–1P. (B refers to blank -nothing loaded in gel; L is a lane used for a ladder).

(C and D) Individual curves of cholesterol levels in fractionated plasma samples from 3m APOE3 and APOE3ch mice shown in the Figure 1Q.

(E and F) Individual curves of APOE levels determined by ELISA in fractionated plasma samples from 3m APOE3 and APOE3ch mice shown in the Figure 1R.

(G and H) Raw Western blot images used for quantification results shown in Figures 2G–2L and 2M–2R. (B refers to blank -nothing loaded in gel; L is a lane used for a ladder).

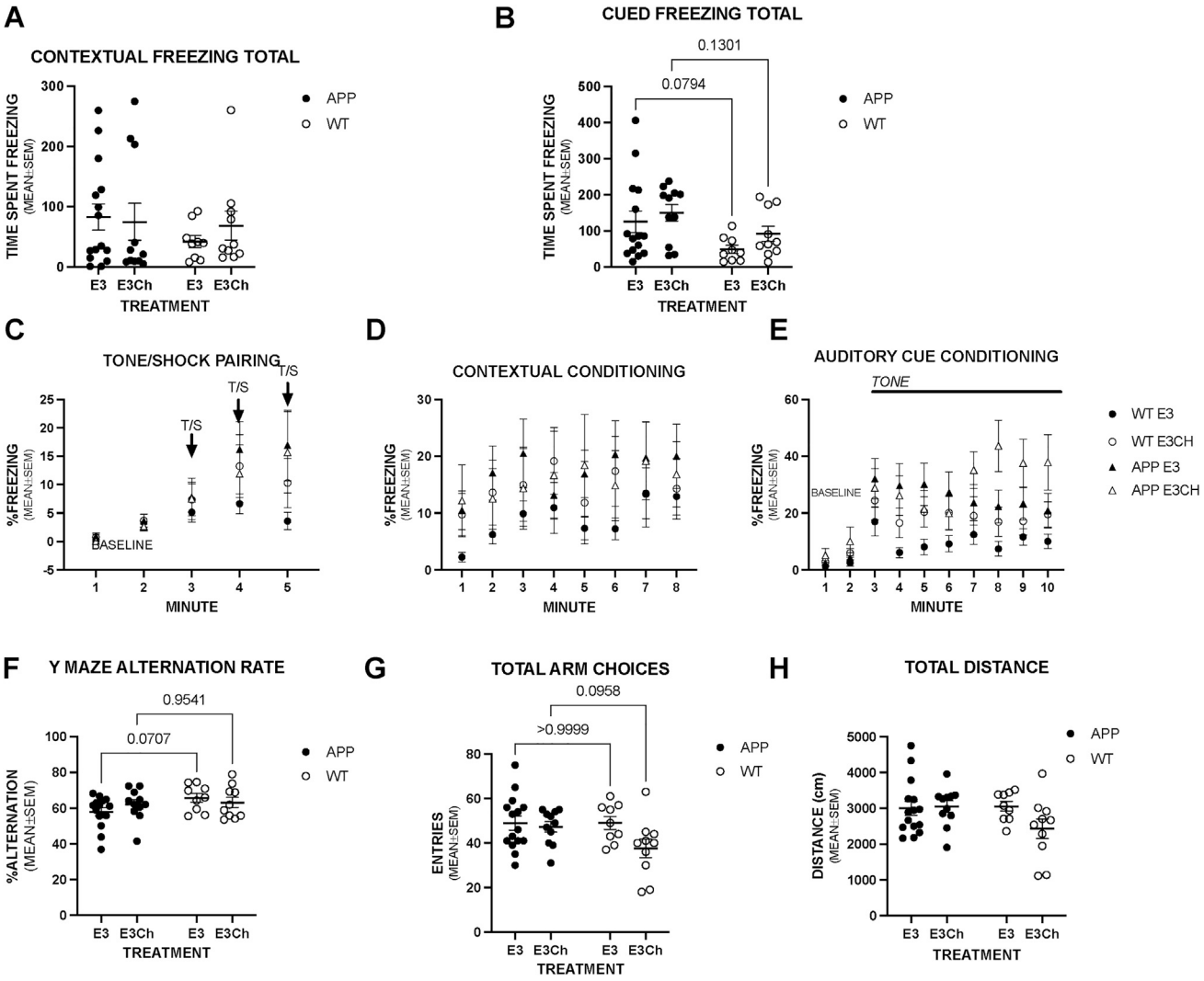


Figure S2. Animal behavior results among APOE3 vs. APOE3ch at 8–10 months and APP/PS1:APOE3 vs. APP/PS1:APOE3ch at 4–6 months, related to Figures 2 and 3

Behavior of 4–6 month old APP/PS1:APOE3 (n = 15) and APP/PS1:APOE3ch (n = 11) mice and 8–10 month old APOE3 (n = 9) and APOE3ch mice (n = 10) in A. contextual freezing test (Total time spend freezing) and B, cued freezing tests (Total time spend freezing).

(C) Tone shocking pairing.

(D) Contextual Conditioning.

(E) Auditory Cued conditioning.

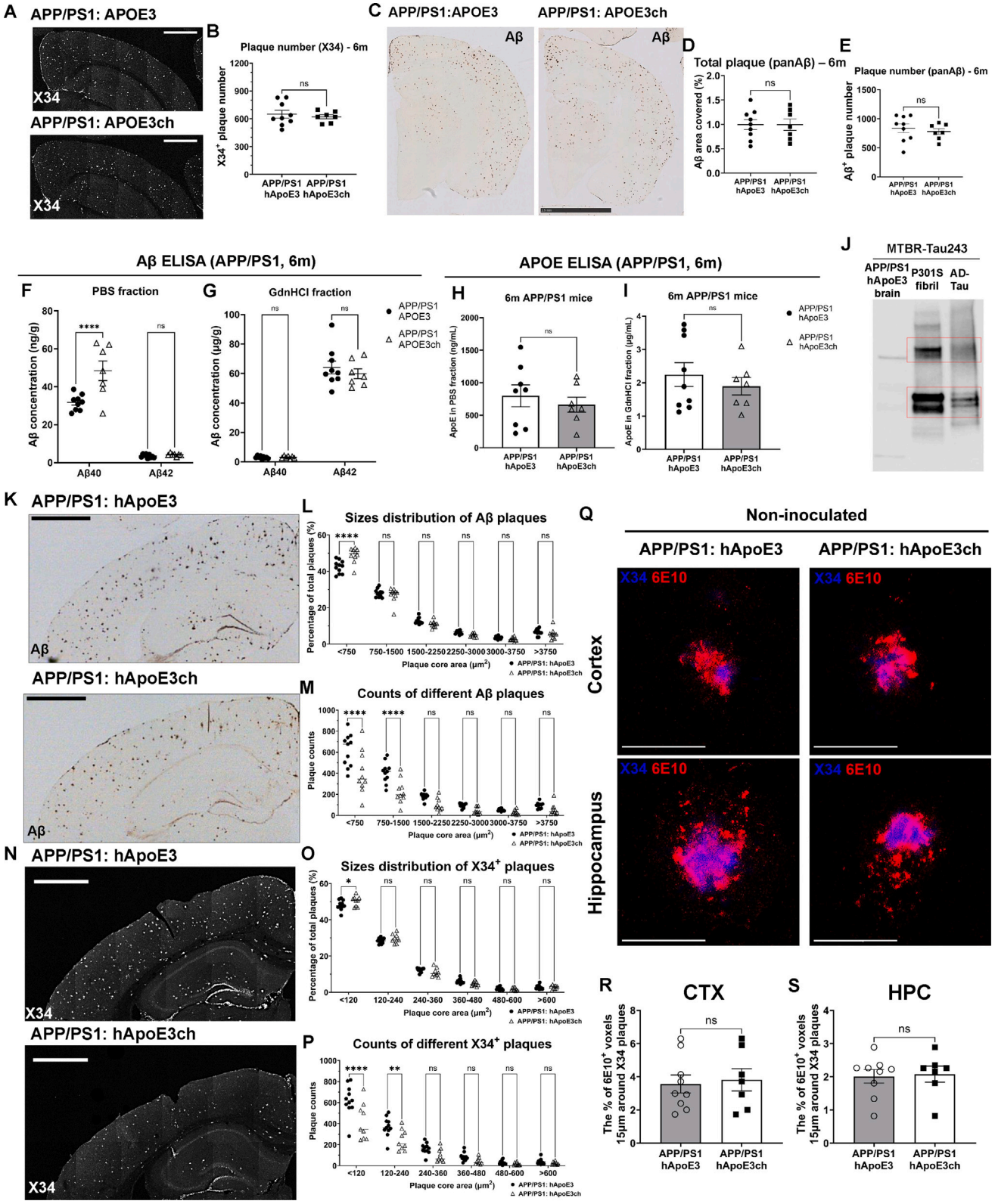
(F) Y maze test alternation rate.

(G) Total Arm choices in the Y-maze.

(H) Total distance traveled in the Y-maze.

For all animal behaviors, statistical analyses were performed using GraphPad Prism (v.10). Analysis of variance (ANOVA), (one-way, factorial, repeated measures) were used analyze the behavioral data. With a statistically significant interaction between main factors, simple main effects were calculated to provide clarification of statistically significant between-genotype and within-genotype differences. Bonferroni correction was applied to multiple pairwise comparisons and post hoc tests. Probability value for all analyses was $p < 0.05$. N = 8/7, 4-6m APP/PS1: APOE3 males/females; N = 5/7, 4-6m APP/PS1: APOE3ch males/females; N = 5/4, 8-10m APOE3 males/females; N = 7/3, 8-10m APOE3ch males/females.

See also Table S1.



(legend on next page)

Figure S3. Additional characterization of amyloid pathology between APP/PS1: APOE3 and APP/PS1: APOE3ch cohorts, related to Figure 3

(A and B) Representative images of X34 staining of Aβ plaques in cortex and hippocampus in non-inoculated APP/PS1:APOE3 and APOE3ch and quantification of plaque numbers. Scale bar in A, 1000μm. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch.

(C–E) Representative images of anti-Aβ (HJ3.4) DAB staining of total Aβ in hemi-brains in non-inoculated APP/PS1:APOE3 and APOE3ch and quantifications as labeled. Scale bar in C, 2.5 mm. CTX, cortex. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch.

(F and G) Aβ40 and Aβ42 ELISA of PBS-soluble and GdnHCl-soluble fractions from brain extracts of non-inoculated 6-month-old APP/PS1:APOE3 and APOE3ch mice. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch.

(H and I) APOE ELISA of PBS-soluble and GdnHCl-soluble fractions from brain extracts of non-inoculated APP/PS1: APOE3 and APOE3ch mice. N = 9, APP/PS1: APOE3; N = 7, APP/PS1: APOE3ch.

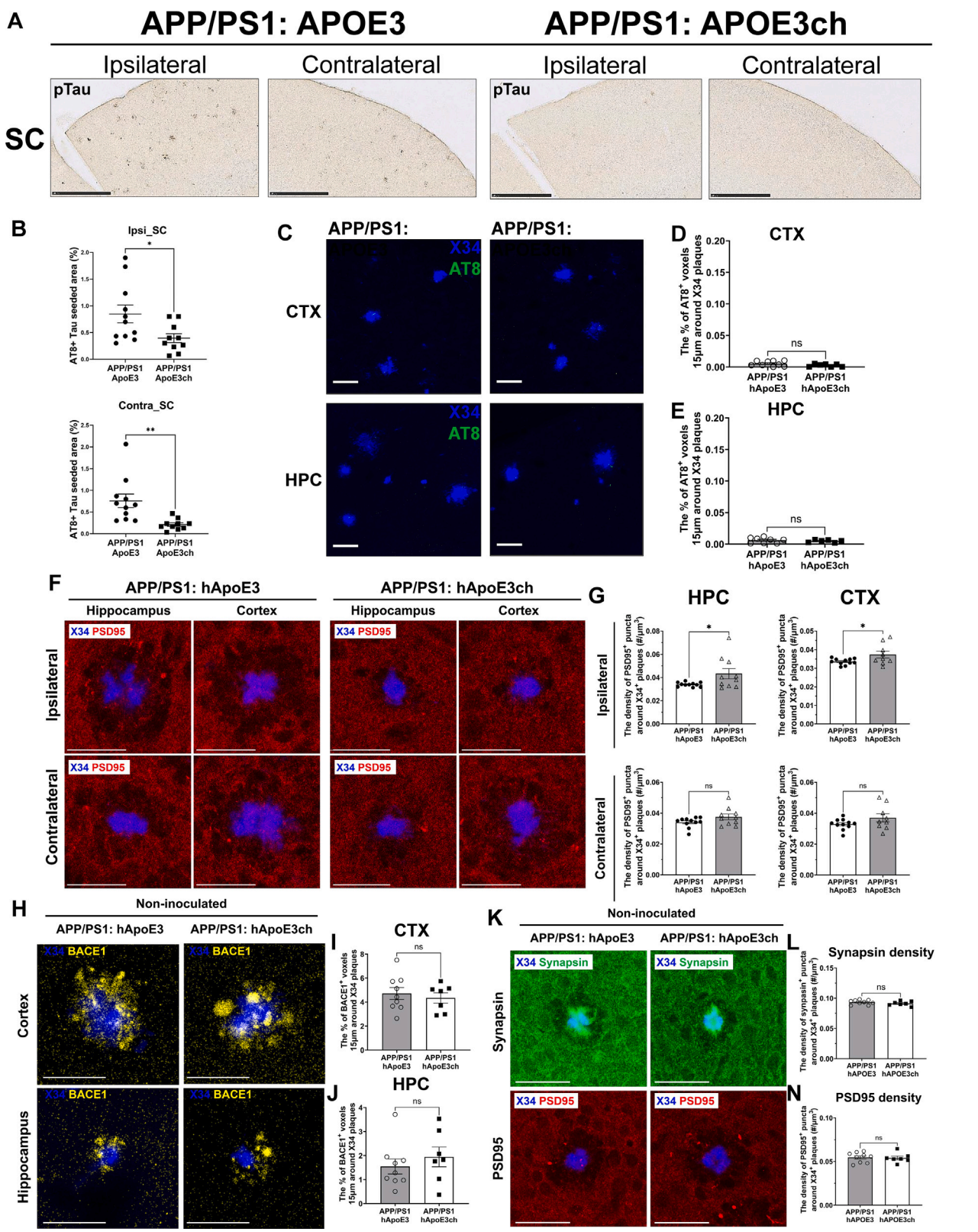
(J) WB image of MTBR-Tau243 probed in the AD-tau extracts as well as APP/PS1: hAPOE brain and from P301S tau fibrils. Red boxes highlight MTBR-Tau243 blotted bands.

(K–M) Representative brightfield images of anti-Aβ staining in the ipsilateral cortex and hippocampus of AD-tau-seeded APP/PS1:APOE3 and APOE3ch and quantification of the percent or number of different areas of Aβ-stained plaques in the whole brain. Scale bars, 1000μm. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(N–P) Representative fluorescence images of X34 staining in the ipsilateral cortex and hippocampus of AD-tau-seeded APP/PS1:APOE3 and APOE3ch and quantification of the percent or number of different areas of Aβ-stained plaques in the whole brain. Scale bars in N, 1000μm. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3.

(Q–S) Representative confocal images of 6E10-647 (anti-Aβ) staining (red) with $X34^+$ plaques (blue) in the APP/PS1:APOE3 and APOE3ch mice and quantification of the percent of $6E10^+$ voxels within 15μm of $X34^+$ plaques. HPC, hippocampus; CTX, cortex. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch. Scale bar in Q, 30 μm.

In all graphs, the statistical analyses were performed with student t-tests, one-way ANOVA or two-way ANOVA and Šídák's multiple comparisons test for averaged values from individual animals. The error bars represent SEM values. The results designated with ns were not significant, whereas results designated with * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$) were significant.



(legend on next page)

Figure S4. Additional characterization of neural dystrophy related pathology in AD-tau seeded APP/PS1:APOE3 vs. APOE3ch mice at 9.5m and in non-inoculated APP/PS1:APOE3 vs. APOE3ch mice at 6m, related to Figure 4

(A and B) Representative images of AT8 DAB staining of pTau (Ser202/Ser205) in AD-tau-seeded APP/PS1:APOE3 and APOE3ch mice and quantification in addition to Figure 4A. Ipsi: ipsilateral; Contra: contralateral. SC: somatosensory cortex. Scale bars in A, 250 μm. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

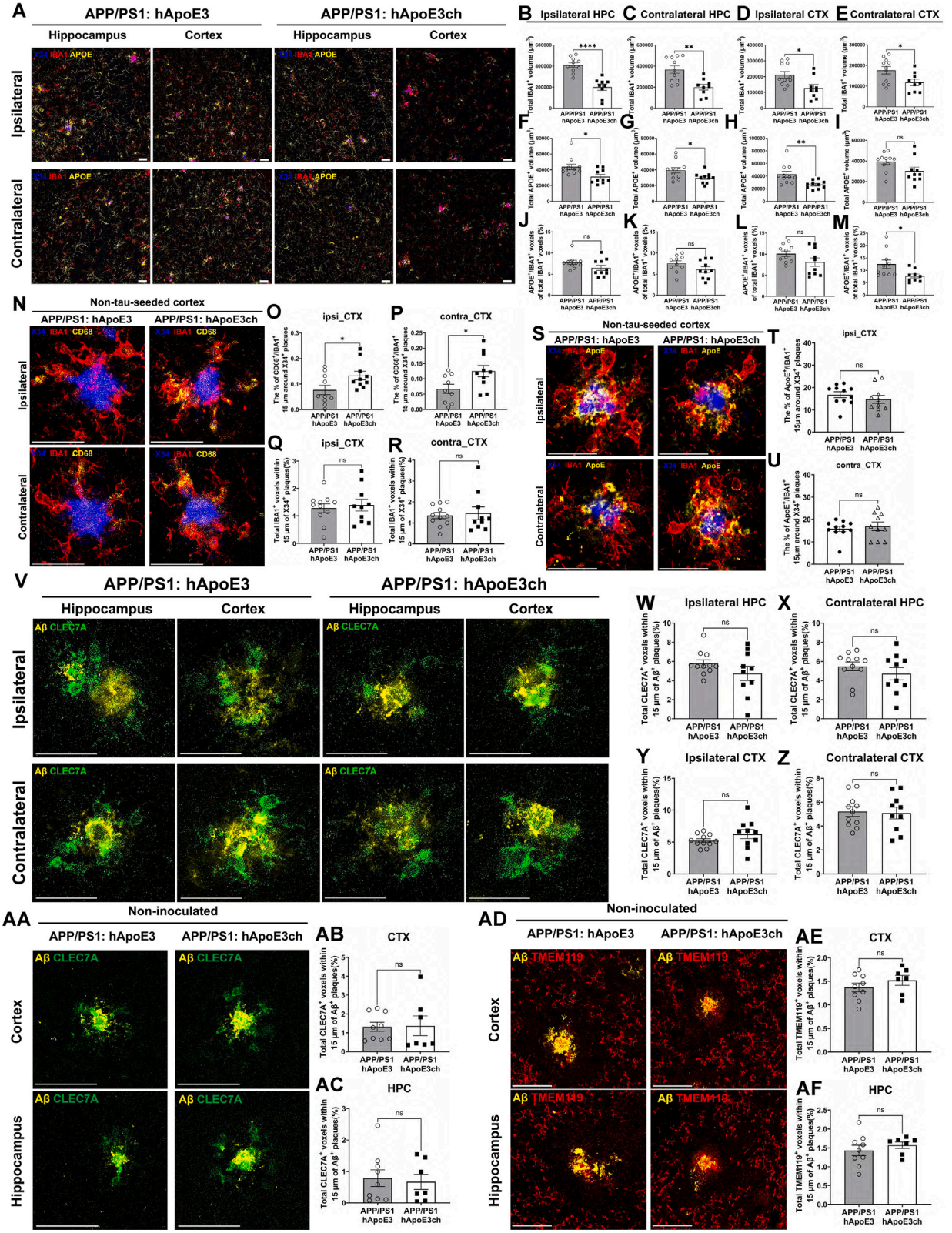
(C–E) Representative confocal images of AT8-labeled (green) NP-tau pathology around $X34^+$ (blue) plaques in the non-inoculated APP/PS1:APOE3 and APOE3ch mice (6m) and quantification. HPC: hippocampus; CTX, cortex. Scale bars in C, 30 μm. N = 9, APP/PS1: APOE3; N = 7, APP/PS1: APOE3ch.

(F and G) Representative confocal images of PSD95-labeled (red) synapses around $X34^+$ (blue) plaques in the AD-tau-seeded APP/PS1:APOE3 and APOE3ch mice and quantification of the count of $PSD95^+$ puncta per μm^3 within 15μm around $X34^+$ plaques. HPC: hippocampus; CTX, cortex. Scale bars in F, 30 μm. N = 11, APP/PS1: APOE3; N = 10, APP/PS1: APOE3ch.

(H–J) Representative confocal images of BACE1-labeled (red) dystrophic neurites around $X34^+$ (blue) plaques in the non-inoculated APP/PS1:APOE3 and APOE3ch mice (6m) and quantification of the percent of $BACE1^+$ voxels within 15μm around $X34^+$ plaques. HPC: hippocampus; CTX, cortex. Scale bars in H, 30 μm. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch.

(K–N) Representative confocal images of synapsin-labeled (green) synapses and PSD95-labeled (red) synapses around $X34^+$ (blue) plaques in the non-inoculated APP/PS1:APOE3 and APOE3ch mice (6m) and quantification of the count of $synapsin^+$ or $PSD95^+$ puncta per μm^3 within 15μm around $X34^+$ plaques. HPC: hippocampus; CTX, cortex. Scale bars in K, 30 μm. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch.

In all graphs, the statistical analyses were performed with student t-tests for averaged values from individual animals. The error bars represent SEM values. The results designated with ns were not significant, whereas results designated with * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$) were significant.



(legend on next page)

Figure S5. Additional characterization of microglial responses in non-inoculated APP/PS1:APOE3 vs. APOE3ch mice at 6m and in AD-tau seeded APP/PS1:APOE3 vs. APOE3ch mice at 9.5m, related to Figure 5

(A) Representative confocal images of IBA1 (yellow), APOE (red) co-stained with X34$^+$ (blue) plaques in the AD-tau-seeded APP/PS1:APOE3 and APOE3ch mice. Scale bars, 30μm.

(B–E) Quantification of the total IBA1$^+$ volumes per image across different regions. HPC: hippocampus; CTX, cortex. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(F–I) Quantification of the total APOE$^+$ volumes per image across different regions. HPC: hippocampus; CTX, cortex. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(J–M) Quantification of the % of APOE$^+$ voxels within IBA1$^+$ voxels across different regions. HPC: hippocampus; CTX, cortex. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(N–P) Representative confocal images of IBA1 (red), CD68 (yellow) co-stained with X34$^+$ (blue) plaques in the non-tau-seeded cortex (anterior) from AD-tau-seeded APP/PS1:APOE3 and APOE3ch mice and quantification of the percent of CD68$^+$/IBA1$^+$ voxels within 15μm around X34+ plaques. Scale bars in N, 30μm. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(Q and R) Quantification of the percent of IBA1$^+$ voxels within 15μm around X34+ plaques in the non-tau-seeded cortex (anterior) from AD-tau-seeded APP/PS1:APOE3 and APOE3ch mice. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

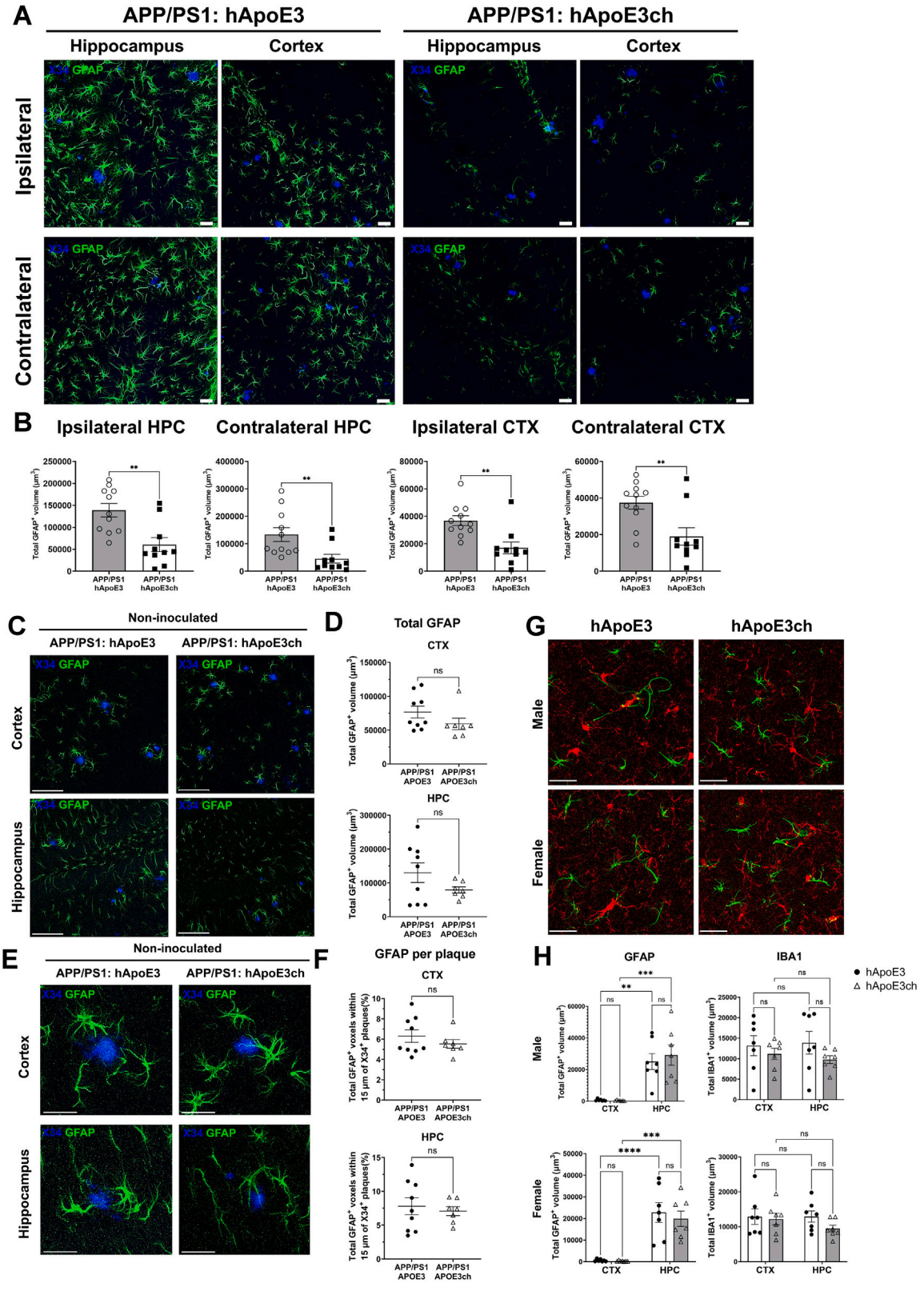
(S–U) Representative confocal images of IBA1 (red), APOE (yellow) co-stained with X34$^+$ (blue) plaques in the non-tau-seeded cortex (anterior) from AD-tau-seeded APP/PS1:APOE3 and APOE3ch mice and quantifications of the percent of APOE$^+$/IBA1$^+$ voxels within 15μm around X34$^+$ plaques. Scale bars in S, 30μm. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(V–Z) Representative confocal images of CLEC7A (green), co-stained with an anti-Aβ antibody HJ3.4$^+$ (yellow) plaques in the AD-tau-seeded APP/PS1:APOE3 and APOE3ch mice and quantification of the percent of CLEC7A$^+$ voxels within 15μm around HJ3.4$^+$ plaques in different regions. Scale bars in V, 30μm. HPC: hippocampus; CTX, cortex. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

(AA–AC) Representative confocal images of CLEC7A (green), co-stained with an anti-Aβ antibody HJ3.4$^+$ (yellow) plaques in the non-inoculated APP/PS1:APOE3 and APOE3ch mice and quantification of the percent of CLEC7A$^+$ voxels within 15μm around HJ3.4$^+$ plaques in different regions. Scale bars in AA, 30μm. HPC: hippocampus; CTX, cortex. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch.

(AD–AF) Representative confocal images of TMEM119 (red) co-stained with HJ3.4$^+$ (yellow) plaques in the non-inoculated APP/PS1:APOE3 and APOE3ch mice and quantification of the percent of TMEM119$^+$ voxels within 15μm around HJ3.4$^+$ plaques. Scale bars in AD, 50 μm. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch.

In all graphs, the statistical analyses were performed with student t-tests for averaged values from individual animals. The error bars represent SEM values. The results designated with ns were not significant, whereas results designated with * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$) were significant. See also Table S1.



(legend on next page)

Figure S6. Reduced total GFAP staining but no difference in astrocyte clustering around plaques in APP/PS1: APOE3ch mice, related to Figures 3, 4, and 5

(A and B) Representative confocal images of GFAP (green) co-stained with $X34^+$ (blue) plaques in the AD-tau-seeded APP/PS1:APOE3 and APOE3ch mice and quantification of the total $GFAP^+$ volumes per image across different regions. Scale bars in A, 30μm. HPC: hippocampus; CTX, cortex. N = 11, APP/PS1:APOE3; N = 10, APP/PS1:APOE3ch.

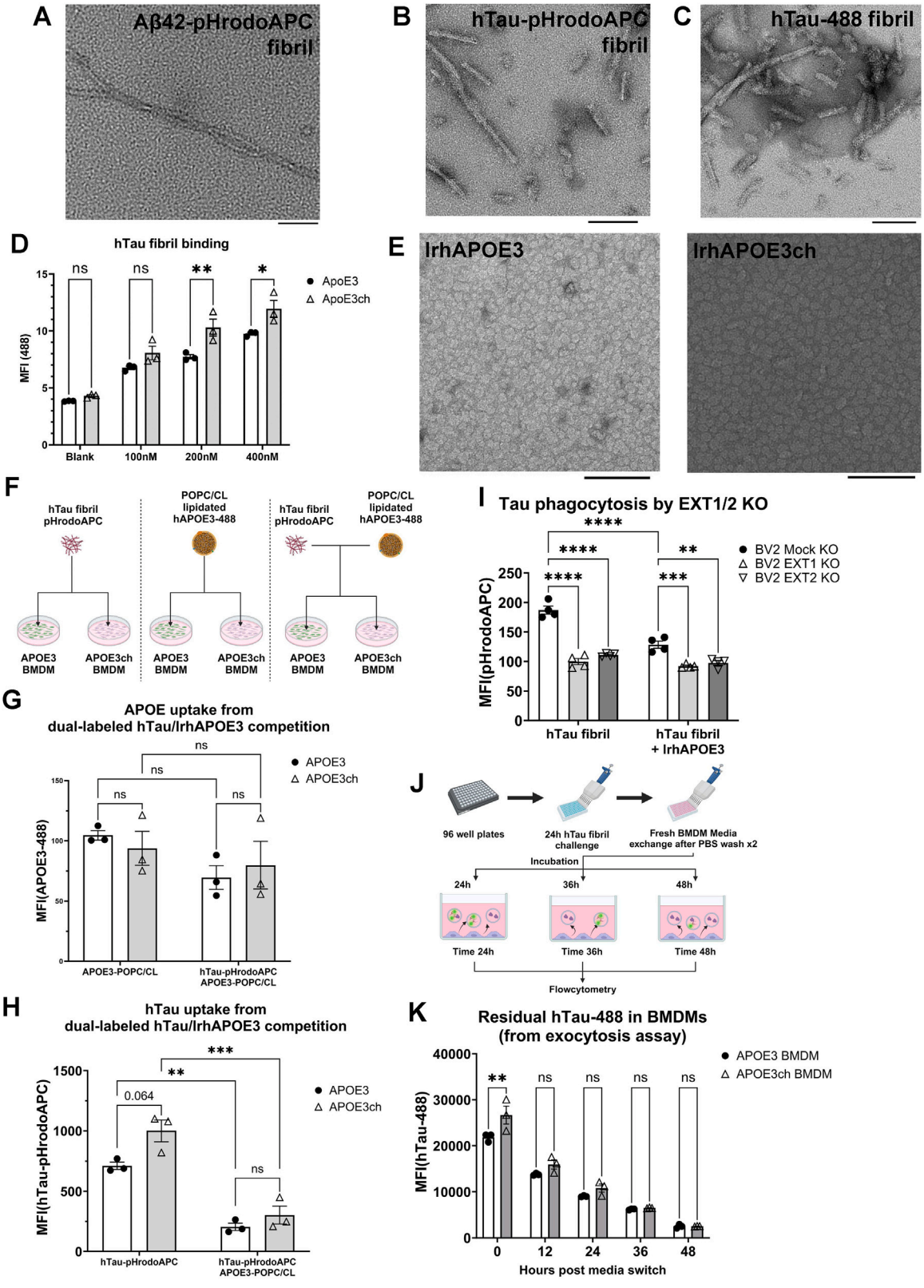
(C and D) Representative confocal images of GFAP (green) co-stained with $X34^+$ (blue) plaques in 6 m APP/PS1:APOE3 and APOE3ch mice and quantification of the total $GFAP^+$ volumes per image. Scale bars in C, 100μm. HPC: hippocampus; CTX, cortex. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch.

(E and F) Zoom-in representative images of GFAP (green) co-stained with $X34^+$ (blue) plaques in 6 m APP/PS1:APOE3 and APOE3ch mice quantification of the percent of $GFAP^+$ voxels within 15μm around $X34^+$ plaques in non-inoculated APP/PS1:APOE3 and APOE3ch mice. HPC: hippocampus; CTX, cortex. N = 9, APP/PS1:APOE3; N = 7, APP/PS1:APOE3ch. Scale bars in E are 30 μm.

(G and H) Representative images of IBA1 (red) and GFAP (green) in 10 m APOE3 and APOE3ch mice and quantifications of the total $IBA1^+/GFAP^+$ volumes per image. N = 7/7, 10m APOE3 males/females; N = 7/7, 10m APOE3ch males/females. Scale bars in G are 30 μm.

The statistical analyses were performed with student t-tests for averaged values from individual animals. The error bars represent SEM values. The results designated with ns were not significant, whereas results designated with * ($p < 0.05$), or ** ($p < 0.01$) were significant.

See also Table S1.



(legend on next page)

Figure S7. Additional *in vitro* experiments in the assessment of APOE3ch protective mechanism, related to Figure 7

(A) The NS-TEM image of Aβ-pHrodoAPC fibrils prepared for *in vitro* experiments. Scale bar: 100 nm.

(B) The NS-TEM image of hTau-pHrodoAPC fibrils prepared for *in vitro* experiments. Scale bar: 100 nm.

(C) The NS-TEM image of hTau-488 fibrils prepared for *in vitro* experiments. Scale bar: 100 nm.

(D) The mean fluorescence intensity (MFI) +/− SEM (Alexa Fluor 488^+) of live BMDMs from each biological replicate incubated with different concentrations of hTau-488 fibrils at 4°C.

(E) The NS-TEM image of POPC+Cholesterol lipidated APOE nanoparticles prepared for *in vitro* experiments. Scale bar is 200 nm.

(F) The experimental design of dual labeled competitive uptake monitor with lrhAPOE3-488 and hTau-pHrodoAPC fibrils.

(G and H) The mean fluorescence intensity (MFI) +/− SEM (Alexa Fluor 488^+ from APOE3 and pHrodoAPC$^+$ from hTau fibrils) of live BMDMs from all experimental replicates with biological triplicates incubated under different conditions.

(I) The mean fluorescence intensity (MFI) +/− SEM (pHrodoAPC$^+$ from hTau fibrils) of live BV2s with EXT1/2 KO from 4 experimental replicates.

(J) The experimental design of assessing exocytotic release of hTau-488 fibrils by BMDMs after hTau phagocytosis.

(K) The mean fluorescence intensity (MFI) +/− SEM (Alexa Fluor 488^+) of live BMDMs from each biological replicates at different timepoints from the exocytosis assay. Related to Figure 7P.

In all BMDM culture experiments, each data point represents the averaged MFI from technical triplicates of one biological replicate BMDM derived from labeled genotype (details in the methods), unless otherwise stated. In all graphs, the statistical analyses were performed with two-way ANOVA and Šídák's multiple comparisons test for the adjusted p values. The error bars represent SEM values. The results designated with ns were not significant, whereas results designated with * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$) or **** ($p < 0.0001$) were significant according to adjusted p values in each indicated comparison.

See also Table S1.